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A Phase 1, First-in-Human, Combined IND/IDE Clinical Trial Protocol

On-Premises LLM-Directed Robotic Pancreaticoduodenectomy (Whipple) with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma

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Protocol version: v4.0.0 **Date:** June 20, 2026 **Regulatory basis:** IND (21 CFR part 312) + IDE (21 CFR part 812), Physical AI Subpart J

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Keywords pancreatic ductal adenocarcinoma; robotic Whipple; daraxonrasib; Physical AI; on-premises LLM; IND/IDE; Phase 1; VVUQ

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1 Statement of Compliance

This trial will be conducted in accordance with the protocol, the principles of the International Council for Harmonisation (ICH) Good Clinical Practice (GCP) guideline E6(R3), and all applicable United States regulations governing both the investigational drug and the investigational device, together with the Physical AI overlay that governs the on-premises large language model (LLM) directed robotic system. Because this is a combined Investigational New Drug (IND) and Investigational Device Exemption (IDE) first-in-human study, two distinct but harmonized regulatory spines apply concurrently; both are carried in full below, and neither is subordinated to the other. The Sponsor-Investigator attests that the conduct of every study activity, the protection of every enrolled participant, and the integrity of every recorded datum will satisfy the more stringent of the two spines wherever they differ.

1.1 Drug Arm Compliance (IND, 21 CFR Part 312)

The perioperative daraxonrasib (RMC-6236) component is conducted under an IND in accordance with Title 21 of the Code of Federal Regulations. Specifically, the study complies with 21 CFR part 50 (Protection of Human Subjects, including the informed-consent requirements of §50.20 through §50.27); 21 CFR part 54 (Financial Disclosure by Clinical Investigators); 21 CFR part 56 (Institutional Review Boards); and 21 CFR part 312 (Investigational New Drug Application), including the sponsor and investigator responsibilities of subparts B and D, the safety-reporting requirements of §312.32, and the IND-application and amendment provisions of §312.20 through §312.31. The drug arm is structured as a Phase 1, 3+3 dose-finding evaluation; the IND is in effect following the 30-day review period of §312.40, and no participant is dosed before the IND is in effect.

1.2 Device Arm Compliance (IDE, 21 CFR Part 812)

The on-premises LLM-directed eight-arm robotic pancreaticoduodenectomy (Whipple) system is an investigational medical device evaluated under an IDE in accordance with 21 CFR part 812. The Sponsor-Investigator and the reviewing Institutional Review Board (IRB) have determined that the device presents a significant risk within the meaning of §812.3(m), because it is a surgical-intervention device whose patient-contact motion, force application, vascular-exclusion gating, and emergency-stop behavior present a potential for serious risk to the health, safety, or welfare of a participant. The study therefore proceeds only under an IDE approved by both the FDA and the IRB. The device arm additionally complies with 21 CFR part 50 and 21 CFR part 56 for human-subject protection and IRB oversight, and with 21 CFR part 11 (Electronic Records; Electronic Signatures) for the validation, audit-trail, and electronic-signature controls that underpin the hash-chained command and telemetry record described in §10. The device component is framed as an early feasibility study consistent with FDA guidance for significant-risk devices, in which limited clinical experience is sought because nonclinical simulation alone cannot supply the information needed to advance development.

1.3 Physical AI Overlay (21 CFR Part 312, Subpart J)

The autonomy-bearing behavior of the LLM-directed system is governed by the Physical AI overlay codified in 21 CFR part 312, Subpart J (§312.400 through §312.405), as adapted for end-to-end Physical AI oncology investigations [1]. Subpart J requires, as preconditions to first-in-human use, a Phase 0 simulation-validation gate (at least two independent simulation frameworks reproducing motion within a ≤ 2 mm positional and ≤ 0.5 N force tolerance), a documented Unified Safety Level (USL) readiness rating of ≥ 7.0 for surgical deployment, and classification of the system as a Class II collaborative device under continuous human oversight with a guaranteed emergency-stop latency of ≤ 500 ms. These overlay requirements operate in addition to, and never in place of, the part 312 and part 812 obligations above. The hash-chained audit trail required under §312.404 satisfies the part 11 electronic-records standard and renders every LLM advisory, every robot command, and every safety-limit event re-traceable to a deterministic seed and a specific repository commit.

1.4 Good Clinical Practice and Electronic Records

The trial adheres to ICH E6(R3) GCP, the international ethical and scientific quality standard for designing, conducting, recording, and reporting clinical investigations involving human participants. All electronic clinical data, source records, and the cyber-physical telemetry stream are maintained under 21 CFR part 11, with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate. The hash-chained record specified by the Physical AI overlay provides cryptographic continuity across the screening, operative, and follow-up phases.

1.5 Trial Registration

The study will be registered on ClinicalTrials.gov before enrollment of the first participant. It is registered as an Interventional study. The drug component is classified as Phase 1 (dose-finding for daraxonrasib), while the device component carries Phase: Not Applicable, consistent with the ClinicalTrials.gov convention that FDA-defined phase categories apply to drugs and biologics rather than to devices. The Primary Purpose is recorded as Device Feasibility together with the dose-finding objective of the drug arm. The study is open-label and single-arm, with staggered (sentinel) enrollment and prespecified pause and stopping rules.

1.6 Statement of Compliance Pathway

Figure 1 renders the combined IND and IDE compliance spine with the Physical AI Subpart J overlay. The daraxonrasib drug arm (IND, part 312) and the robotic Whipple device arm (IDE, part 812, significant-risk) converge through the Phase 0 simulation-validation gate, the USL readiness verification, and the Class II collaborative classification into a single first-in-human protocol.

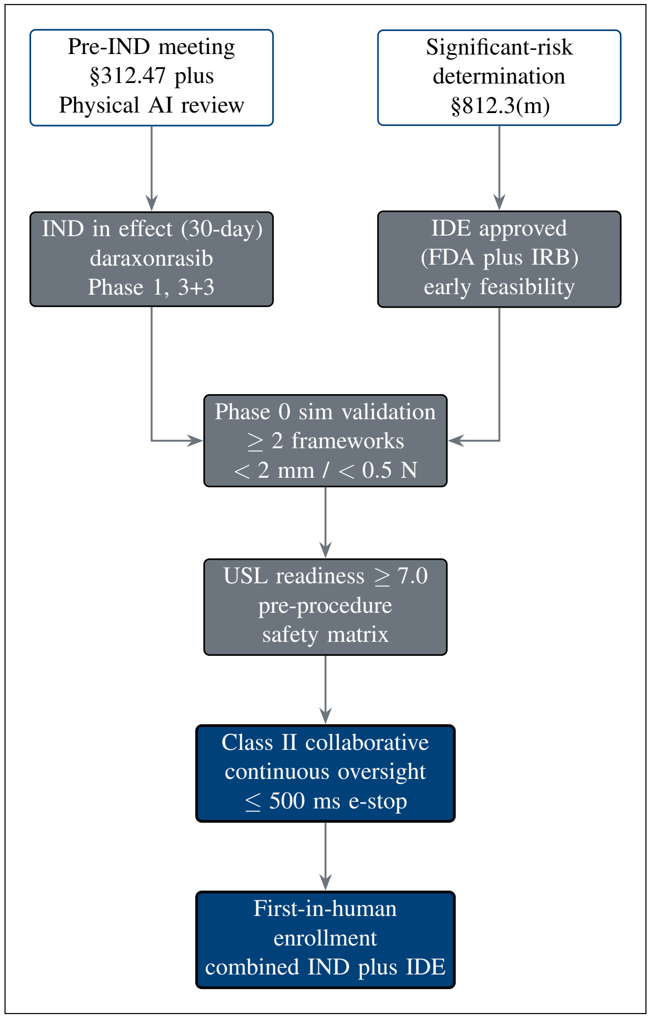

Figure 1. Combined IND / IDE regulatory pathway with the Physical AI overlay. The daraxonrasib drug arm (IND, part 312) and the robotic Whipple device arm (IDE, part 812, significant-risk per §812.3(m)) converge through the Subpart J overlay of Phase 0 simulation validation, USL readiness, and Class II collaborative classification into one first-in-human protocol.

1.7 Signatures

The Sponsor-Investigator has reviewed and approved this protocol and confirms that the study will be conducted in compliance with the protocol, ICH E6(R3) GCP, and the federal regulations cited above. By signature below, each signatory agrees to conduct or supervise the study in accordance with the protocol and all applicable requirements.

Sponsor-Investigator	**Site Principal Investigator**
Kevin Kawchak CEO, ChemicalQDevice	Name (printed) Title / Institution
Signature and date: ______________	Signature and date: ______________

The signatory acknowledges that this is an independent research draft; it is not an active IND or IDE and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

2 Protocol Summary

2.1 Synopsis

Title. A Phase 1, first-in-human, combined IND/IDE clinical trial of on-premises large language model (LLM) directed robotic pancreaticoduodenectomy (Whipple) with perioperative daraxonrasib (RMC-6236) in KRAS-mutated pancreatic ductal adenocarcinoma (PDAC).

Rationale. PDAC is the third leading cause of cancer death in the United States, with total five-year overall survival below 13 percent [2], and the Whipple procedure is the most technically demanding of all curative-intent abdominal oncology operations. This study pairs a precise, auditable, on-premises LLM-directed robotic Whipple with the RAS(ON) multi-selective inhibitor daraxonrasib, with the aim of achieving an in-window R0 resection while layering hard cyber-physical safety limits and optimizing the perioperative drug-restart window.

Objectives and endpoints. The primary objective is to characterize the safety and feasibility of the combined intervention. The primary endpoints are the rate of dose-limiting toxicities (DLTs) during the daraxonrasib 3+3 escalation and the rate of device-related and procedure-related serious adverse events through the day-30 safety window, including Clavien-Dindo grade III or higher complications. Key secondary endpoints are the R0 resection rate at day-90 pathology, the International Study Group on Pancreatic Surgery (ISGPS) grade B/C postoperative pancreatic fistula rate, 90-day mortality, progression-free survival (PFS), and overall survival (OS) to 24 months.

Design. Open-label, single-arm, dose-finding study using a standard 3+3 escalation across three daraxonrasib dose levels (DL1 through DL3) with staggered (sentinel) enrollment. Enrollment is planned for up to $n = 18$ participants. A mandatory Phase 0 simulation-validation gate (≥ 1000 simulated procedures across ≥ 2 independent frameworks, with a Unified Safety Level (USL) rating ≥ 7.0) precedes any patient-facing robotic activity, and prespecified pause and stopping rules govern progression.

Population. Adults (≥ 18 years) with histologically confirmed, resectable or borderline-resectable, KRAS G12-mutated PDAC; ECOG performance status 0 or 1; and adequate organ function. Participants are daraxonrasib-eligible under the broad pan-KRAS criteria of the RASolute 302 program [3]. Full inclusion and exclusion criteria appear in §5.

Intervention. Perioperative daraxonrasib administered with a planned preoperative pause and an LLM-advised postoperative restart at T+7, T+14, or T+21 days keyed to fistula status and drug trough, combined with an eight-arm robotic Whipple performed under continuous human oversight as a Class II collaborative device, with a 10 kHz heartbeat broadcast bus, a ≤ 3 ms cross-arm emergency-stop budget, per-arm tip-force caps of ≤ 3 N, a cumulative cross-arm tip-force cap of ≤ 18 N, and vascular no-fly gating around the superior mesenteric and portal veins.

Duration. Each participant undergoes up to 28 days of screening, a baseline day, surgery on day 0, an acute day 1 to 7 window, the day-30 primary safety window, day-90 pathology and mortality assessment, and long-term follow-up every 12 weeks to the 24-month overall-survival endpoint.

2.2 Schema

Figure 2 presents the end-to-end trial schema, from 28-day screening through enrollment with the Physical AI consent opt-out, the Phase 0 simulation-validation gate, daraxonrasib 3+3 dose assignment, the eight-arm robotic Whipple, the perioperative pause-and-restart advisory, follow-up, and the 24-month overall-survival endpoint.

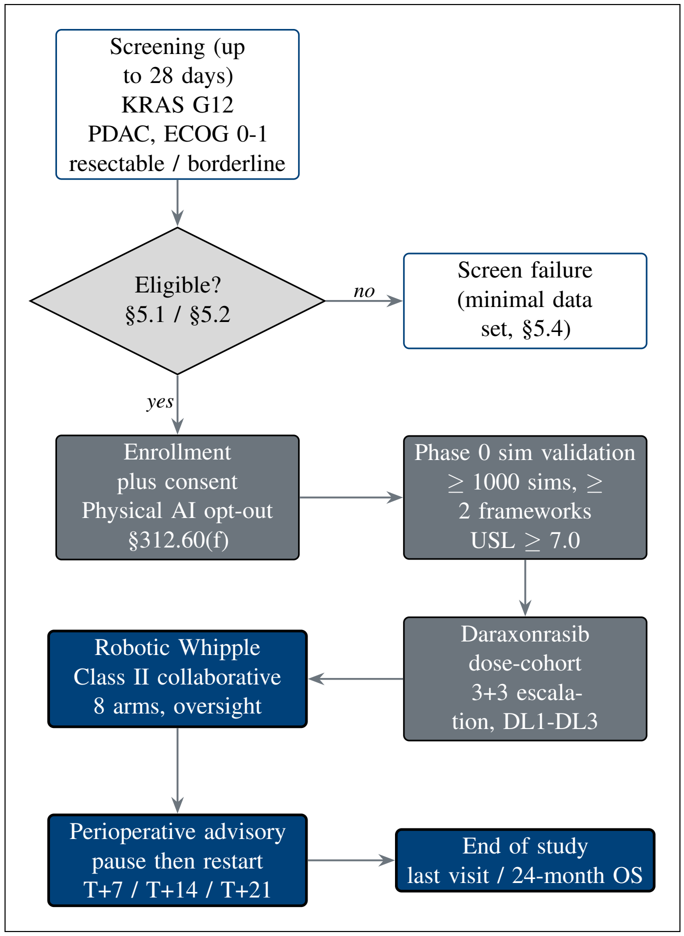

Figure 2. Overall trial schema for the Phase 1 combined IND/IDE study, from 28-day screening through enrollment with the Physical AI consent opt-out, Phase 0 simulation validation, daraxonrasib 3+3 dose assignment, the eight-arm robotic Whipple, the perioperative pause-and-restart advisory, and follow-up to the 24-month overall-survival endpoint.

2.3 Schedule of Activities (SoA)

The Schedule of Activities binds each assessment to a timepoint across the screening window (Day -28 to -1), the baseline day (Day -1), surgery (Day 0), the acute window (Day 1 to 7), Day 30, Day 90, and long-term follow-up every 12 weeks. An "X" marks each assessment performed at that visit. The baseline day carries the Phase 0 simulation sign-off and the USL plus safety-matrix verification, the acute window carries the ISGPS fistula grading and the daraxonrasib restart advisory, and the long-term column carries PFS and OS to 24 months.

Activity	**Screen Day** -28 **to** -1	**Base Day** -1	**Surg Day 0**	**Acute Day 1-7**	**Day 30**	**Day 90**	**LT q12wk**
Informed consent (incl. Physical AI opt-out)	X						
Staging / KRAS G12 confirmation	X						
CA 19-9	X	X			X	X	X
ECOG performance status	X	X			X	X	X
Safety laboratory panel	X	X	X	X	X	X	X
Phase 0 simulation sign-off		X					
USL $\geq$ 7.0 plus safety matrix		X					
Robotic Whipple surgery			X				
Intra-operative telemetry capture			X				
ISGPS fistula grading				X	X		
Daraxonrasib restart advisory				X			
AE / SAE review	X	X	X	X	X	X	X
RECIST imaging	X					X	X
PFS / OS assessment						X	X

Abbreviations: AE, adverse event; SAE, serious adverse event; ISGPS, International Study Group on Pancreatic Surgery; LT, long-term; OS, overall survival; PFS, progression-free survival; RECIST, Response Evaluation Criteria in Solid Tumors; USL, Unified Safety Level. The Day 30 visit is the primary safety window (Clavien-Dindo grading); Day 90 carries 90-day mortality and R0 pathology.

3 Introduction

3.1 Study Rationale

Pancreatic ductal adenocarcinoma (PDAC) is the third leading cause of cancer death in the United States and the fourth in the European Union, with total five-year overall survival below 13 percent [2]. The pancreaticoduodenectomy, or Whipple procedure, is the most technically demanding of all curative-intent abdominal oncology operations: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around two named vessels (the superior mesenteric vein and the portal vein), with an artery-first approach to the superior mesenteric artery. The leading lethal postoperative sequence is the fistula-sepsis cascade: pancreatic fistula, intra-abdominal infection, hemorrhage, sepsis, and failure to rescue, graded under the ISGPS classification of postoperative pancreatic fistula [4]. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, ISGPS grade B or C fistula of 24.4 percent, and 30-day mortality of 3.9 percent [5]; these figures anchor the contemporary human baseline. The present protocol pairs a precise, auditable, on-premises large language model (LLM) directed robotic Whipple with the RAS(ON) multi-selective inhibitor daraxonrasib in patients who have completed or are eligible for systemic therapy such as modified FOLFIRINOX [6].

The clinical argument for the combination is best stated through three counterfactual scenarios, summarized in Figure 3, in which withholding the LLM-plus-robot-plus-medicine combination shortens both PFS and OS. In Scenario A, a borderline-resectable tumor abutting the superior mesenteric vein at 75 degrees progresses to unresectable disease during a human-only scheduling delay (operating room access, fatigue, staging lag), and the R0 window closes; the LLM-directed robot instead achieves a rapid, precise, in-window R0 resection. In Scenario B, an unrecognized superior-mesenteric or portal-vein injury during manual surgery triggers the hemorrhage-conversion-fistula-sepsis-failure-to-rescue cascade; the no-fly vascular gate with a ≤ 3 ms emergency stop averts the injury and preserves a complete resection. In Scenario C, manual mistiming of the daraxonrasib restart (too early or too late) causes anastomotic dehiscence or micrometastatic outgrowth; the LLM advisory restart at T+7, T+14, or T+21 days, keyed to fistula status and drug trough, preserves the therapeutic window.

These scenarios invite three framing questions that this protocol is designed to answer constructively rather than rhetorically. First, given the narrow resection window and the unforgiving fistula-sepsis cascade, how did we ever rely on humans alone for this treatment, with no second-opinion oracle isolated from the operative kinematics. Second, why did we fully trust a prior trial system with so many opportunities for compounding human error across scheduling, dissection, and drug timing, when each step admits a documented, deterministic safeguard. Third, as AI outputs grow more complex and voluminous, why would human peer review alone continue to suffice for increasingly intricate AI-directed care, rather than a verification-before-generation discipline with a hash-chained audit trail [7]. The protocol answers each question by retaining the human in the loop while making every machine action validated, bounded, and re-traceable.

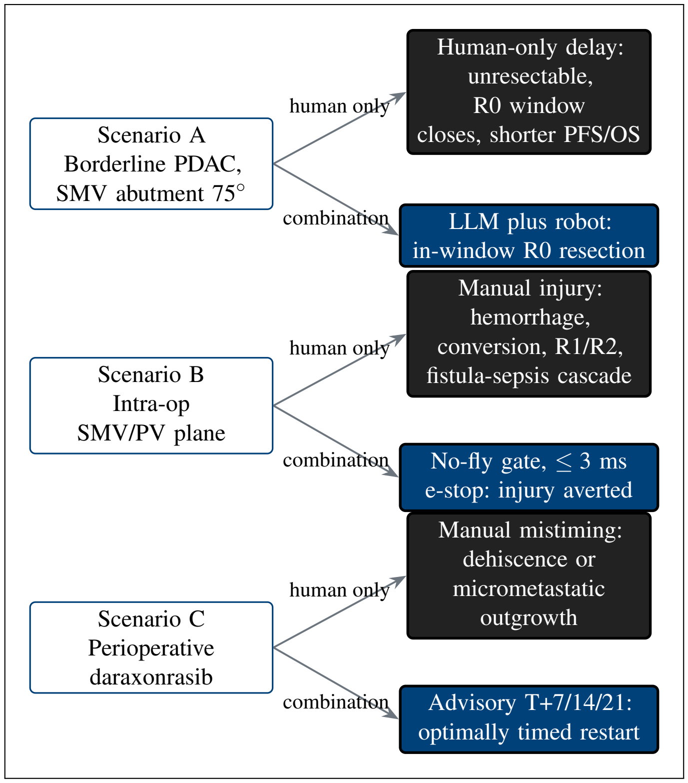

Figure 3. The three counterfactual scenarios in which withholding the combination worsens the patient. Scenario A (top): a human-only scheduling delay lets a borderline-resectable tumor progress and closes the R0 window. Scenario B (middle): an unrecognized superior-mesenteric or portal-vein injury during manual surgery triggers the fistula-sepsis cascade; the no-fly gate with a ≤ 3 ms emergency stop averts it. Scenario C (bottom): manual mistiming of the daraxonrasib restart causes dehiscence or micrometastatic outgrowth, which the timed advisory prevents. In each, the human-only path shortens progression-free and overall survival while the LLM-plus-robot-plus-medicine path preserves them.

3.2 Background

PDAC epidemiology and the Whipple fistula-sepsis cascade

The epidemiologic burden and operative hazard of PDAC define the unmet need this protocol addresses. With under-13-percent five-year survival [2] and a most-demanding curative operation whose leading complication is a self-amplifying fistula-sepsis-hemorrhage cascade [4], the population has few alternatives and a narrow margin for technical error. The contemporary robotic baseline [5] demonstrates that even expert, high-volume robotic surgery leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae. The combination in this protocol targets precisely the failure modes that the baseline still tolerates.

Daraxonrasib mechanism and clinical program

Daraxonrasib (RMC-6236) is an oral RAS(ON) multi-selective inhibitor that binds the active, GTP-bound (ON) state of multiple RAS variants, providing broad pan-KRAS coverage including the G12 alleles that predominate in PDAC. The FDA granted daraxonrasib Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC with KRAS G12 mutations [8], and the late-phase program comprises RASolute 302 (a Phase 3 study in previously treated metastatic PDAC) and RASolve 301 (a first-line program) [3]. The broad pan-KRAS eligibility of RASolute 302 defines the molecular eligibility used in this protocol, and the perioperative pause-and-restart logic respects any RASolve 301 checkpoint-inhibitor combination context. Five prior computational works by the author independently complemented this drug-development program throughout 2025 and 2026 and supply the simulation scaffolding adapted here: a 60-second PDAC robotic Whipple and daraxonrasib simulation [9]; an AI digital-twin pancreatic-cancer simulation for in silico FDA-compliance and cost efficiency [10]; a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with code, VVUQ, and playbook [11]; a 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [12]; and end-to-end PDAC digital-twin clinical-trial proposals [13].

Current state of AI in clinical trials and surgery

Artificial intelligence in regulated medicine has, to date, been narrow and prediction-oriented rather than generative and intervention-directing. As of the current taxonomy of FDA-authorized medical devices, AI is overwhelmingly deployed as prediction-only or narrow task-specific software, with reported authorizations spanning 1,016 in the published taxonomy [14] and rising to 1,450 or more by current counts, while zero generative LLM devices have been approved for patient-facing intervention. This protocol therefore does not propose a generative device; it proposes a narrow, locked, validated device behavior governed by an on-premises repository-based LLM whose advisories are bounded, auditable, and held outside the robot vendor kinematic stack. The author's established record of LLM evaluation and trust over the period from August 2024 to June 2026 grounds this design, including a comparative code-generation evaluation of 16 proprietary versus open-source LLMs against the FDA Adverse Event Reporting System [15], a mobile pancreatic-cancer surgical-humanoid program with priority VVUQ [16], the 21 CFR part 312 Physical AI unification adaptation [1], and the verification-before-generation legislative framing [7].

Physical AI concerns addressed by this protocol

Eight recurring concerns are raised about Physical AI in patient care, and this protocol pairs each with a concrete, prespecified mitigation. Figure 4 maps the eight concerns to their mitigations, and Table 1 states each pairing in full.

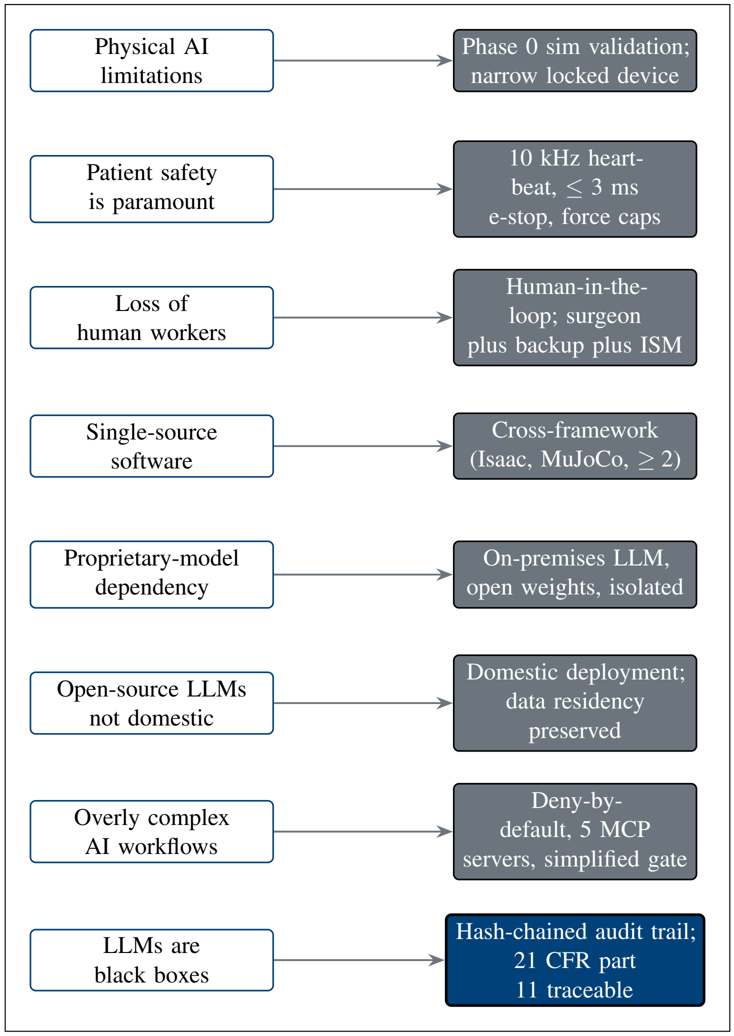

Figure 4. The complete mapping of all eight Physical AI concerns to their protocol mitigations, the same eight pairs tabulated in Table 1. Each concern is paired with a concrete, prespecified safeguard: validated narrow device behavior over generative open-endedness, layered hard safety limits, retained human roles, and multi-framework non-single-source validation.

Concern	Protocol mitigation
Physical AI limitations	Phase 0 simulation validation over ≥ 1000 procedures and ≥ 2 independent frameworks; the device behavior is narrow and locked, not generative, and operates only within validated envelopes.
Patient safety is paramount	Layered hard limits: a 10 kHz heartbeat broadcast bus, a ≤ 3 ms cross-arm emergency stop, ≤ 3 N per-arm and ≤ 18 N cumulative tip-force caps, and vascular no-fly gating around the SMV and PV.
Loss of human workers	The human is retained in the loop: a credentialed surgeon, a backup operator, and an independent safety monitor (ISM) supervise every procedure, with the LLM acting as a bounded advisory oracle rather than a replacement.
Single-source software	Validation is cross-framework (for example Isaac, MuJoCo, Gazebo, PyBullet), with ≥ 2 independent frameworks required, so no single simulator or codebase is a single point of failure.
Proprietary-model dependency	The advisory uses an on-premises repository-based LLM with open weights, isolated from any single vendor stack, so the protocol is not captive to one proprietary model provider [15].
Non-domestic open-source LLMs	Deployment is domestic and on-premises, preserving data residency and avoiding reliance on non-domestic open-source model hosting; inference and records remain within the controlled environment.
Overly complex AI workflows	A deny-by-default posture with a deliberately simplified gate surface (a small set of Model Context Protocol servers) constrains the action space and reduces workflow complexity to an auditable minimum.
LLMs are black boxes	A hash-chained audit trail under 21 CFR part 11 records every advisory, command, and safety event, making each re-traceable to a deterministic seed and a specific commit, so the black box becomes traceable [1].

Table 1. The eight Physical AI concerns and their protocol mitigations.

3.3 Risk/Benefit Assessment

3.3.1 Known Potential Risks

The known potential risks fall into three categories. Drug risks arise from daraxonrasib toxicity, including gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events, hepatic enzyme elevations, and dose-limiting toxicities that the 3+3 escalation is specifically designed to detect and bound; perioperative exposure additionally raises a theoretical risk to wound and anastomotic healing if restart timing is suboptimal. Device and robotic risks include device malfunction, an erroneous AI advisory, cyber-physical failure of the control or telemetry chain, and inadvertent vascular injury; these are the categories that motivate the significant-risk determination and the layered hard safety limits. Operative risks are those intrinsic to the Whipple procedure: postoperative pancreatic fistula (graded by ISGPS [4]), hemorrhage, delayed gastric emptying, intra-abdominal infection and sepsis, and the need for conversion to open surgery. The contemporary robotic baseline quantifies the residual magnitude of these operative risks even in expert hands [5].

3.3.2 Known Potential Benefits

The known potential benefits are directed at the specific failure modes above. A precise, in-window R0 resection is the principal oncologic benefit, addressing the resection-window collapse and the margin-positivity that drive shorter survival. Layered hard safety limits with a full hash-chained audit trail provide a benefit not available in manual surgery: every machine action is bounded and re-traceable, supporting both intraoperative safety and post hoc verification. Optimized drug-restart timing, advised at T+7, T+14, or T+21 days and keyed to fistula status and drug trough, aims to restore systemic therapy at the earliest safe point and thereby suppress micrometastatic outgrowth. For a low-survival population with few alternatives, even incremental gains in margin status, complication avoidance, and therapy continuity are clinically meaningful.

3.3.3 Assessment of Potential Risks and Benefits

Figure 5 frames the benefit-risk weighing for this population. Against a baseline of under-13-percent five-year survival and few alternatives, the novel risks (device malfunction, advisory error, cyber-physical failure) are weighed against the benefits (precise in-window R0 resection, layered hard safety limits with a full audit trail, and optimized drug-restart timing). For this low-survival, limited-alternative population, the benefit-risk balance is favorable: the potential benefits are likely to outweigh the risks, and the assessment is consistent with the expanded-access ethic that permits investigational treatment for serious or immediately life-threatening conditions when the potential benefit justifies the potential risk (21 CFR §312.84).

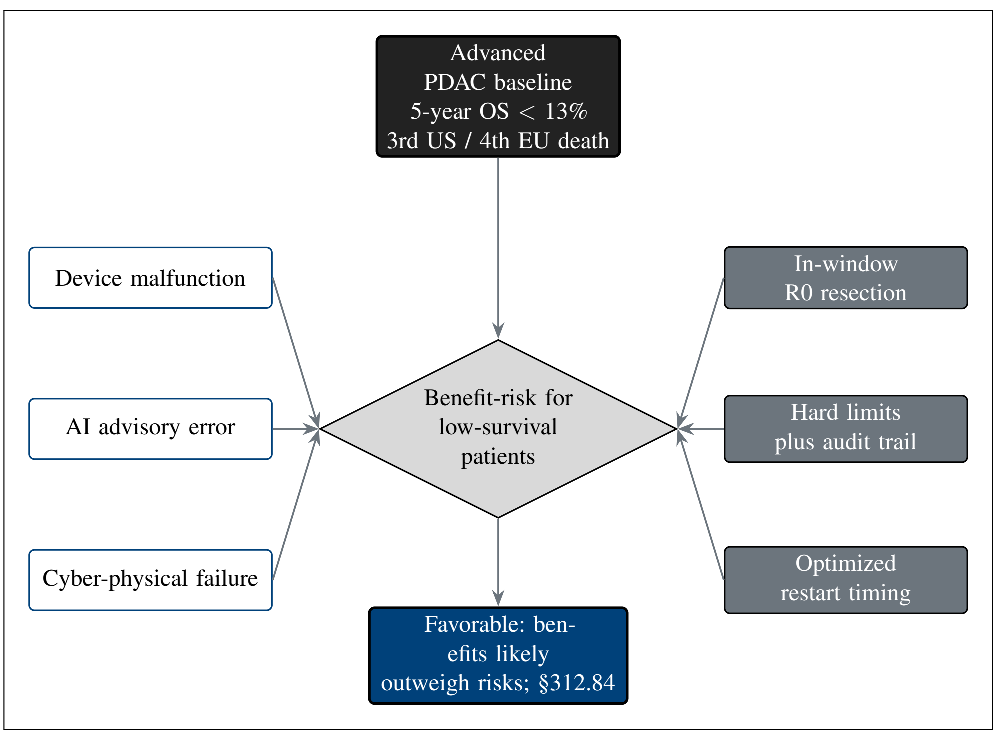

Figure 5. Risk-benefit framework for advanced PDAC. Against a baseline of under-13-percent five-year survival and few alternatives, the novel risks (device malfunction, advisory error, cyber-physical failure) are weighed against the benefits (precise in-window R0 resection, layered hard safety limits with a full audit trail, optimized drug timing). For this low-survival, limited-alternative population the benefit-risk balance is favorable, consistent with the expanded-access ethic of 21 CFR §312.84.

4 Objectives and Endpoints

This study pairs each objective with a prospectively defined, measurable endpoint and a scientific justification, presented in the standard three-column form. The trial is a combined investigation of two regulated articles: perioperative daraxonrasib (RMC-6236) under the investigational new drug (IND) pathway, and the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system under the investigational device exemption (IDE) pathway. The objective set is therefore organized so that the drug, the device, and the combined procedure each carry their own primary safety, dose-finding, and feasibility anchors, with secondary oncologic-surgical quality measures and exploratory survival and human-factors measures layered beneath them. Table 1 is the governing objectives-endpoints table; it drives the analysis plan in §9 and the safety oversight rules in §6 and §7.

Objective	**Endpoint**	**Justification**
Primary		
Characterize the early safety of the combined LLM-directed robotic Whipple and perioperative daraxonrasib intervention.	Incidence of device-related or procedure-related serious adverse events (SAEs) through 30 days after the procedure, including Clavien-Dindo grade III or higher complications.	Thirty-day SAE incidence is the standard first-in-human safety anchor and captures the dominant early post-Whipple cascade (fistula, hemorrhage, sepsis, failure to rescue); a device or procedure attribution isolates risk introduced by the investigational system.
Determine the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D) of perioperative daraxonrasib.	MTD and RP2D defined by the rate of dose-limiting toxicities (DLTs) within the 28-day DLT window of the 3+3 escalation across dose levels DL1 through DL3.	A 3+3 escalation is the established dose-finding design for cytotoxic and targeted oncology agents and yields a defensible RP2D from a small first-in-human cohort, consistent with the RASolute 302 pan-KRAS program [3].
Establish the technical feasibility of the eight-arm robotic Whipple under continuous human oversight.	Proportion of procedures that complete all prespecified assigned operative tasks without unplanned conversion caused by device malfunction or unsafe device behavior.	Task-completion without unsafe conversion is the FDA-recognized early-feasibility readout for a significant-risk device and measures whether the system can perform its intended function within hard cyber-physical limits [17].
Secondary		
Estimate the oncologic adequacy of the resection.	R0 (microscopically margin-negative) resection rate determined at day-90 pathology review.	R0 status is the strongest modifiable surgical determinant of survival in resectable PDAC and is the principal oncologic quality measure for the operation.
Estimate the rate of clinically relevant postoperative pancreatic fistula.	International Study Group on Pancreatic Surgery (ISGPS) grade B or C postoperative pancreatic fistula rate [4]; human comparator 24.4% [5].	Grade B/C fistula is the leading driver of post-Whipple morbidity and mortality; the 2025 Dutch nationwide robotic cohort provides a contemporaneous human benchmark.
Estimate the rate of major postoperative complications.	Clavien-Dindo grade III or higher complication rate through 90 days; human comparator 41.3% [5].	Clavien-Dindo grade III+ is the validated severity threshold for interventions requiring procedural, endoscopic, or intensive-care management.
Estimate early postoperative	30-day and 90-day all-cause	Thirty- and 90-day mortality are

Abbreviations: DLT, dose-limiting toxicity; ISGPS, International Study Group on Pancreatic Surgery; LLM, large language model; MTD, maximum tolerated dose; OS, overall survival; PFS, progression-free survival; RP2D, recommended Phase 2 dose; SAE, serious adverse event. Human comparator values are drawn from the 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies [5] and are descriptive benchmarks only; this single-arm first-in-human study is not powered for formal comparison against them.

Figure 3 renders the objective-to-endpoint hierarchy that organizes Table 1, showing how the three primary anchors (30-day serious adverse event incidence, daraxonrasib MTD/RP2D, and unsafe-conversion-free task completion) sit above the secondary oncologic-surgical quality measures and the exploratory survival and human-factors measures.

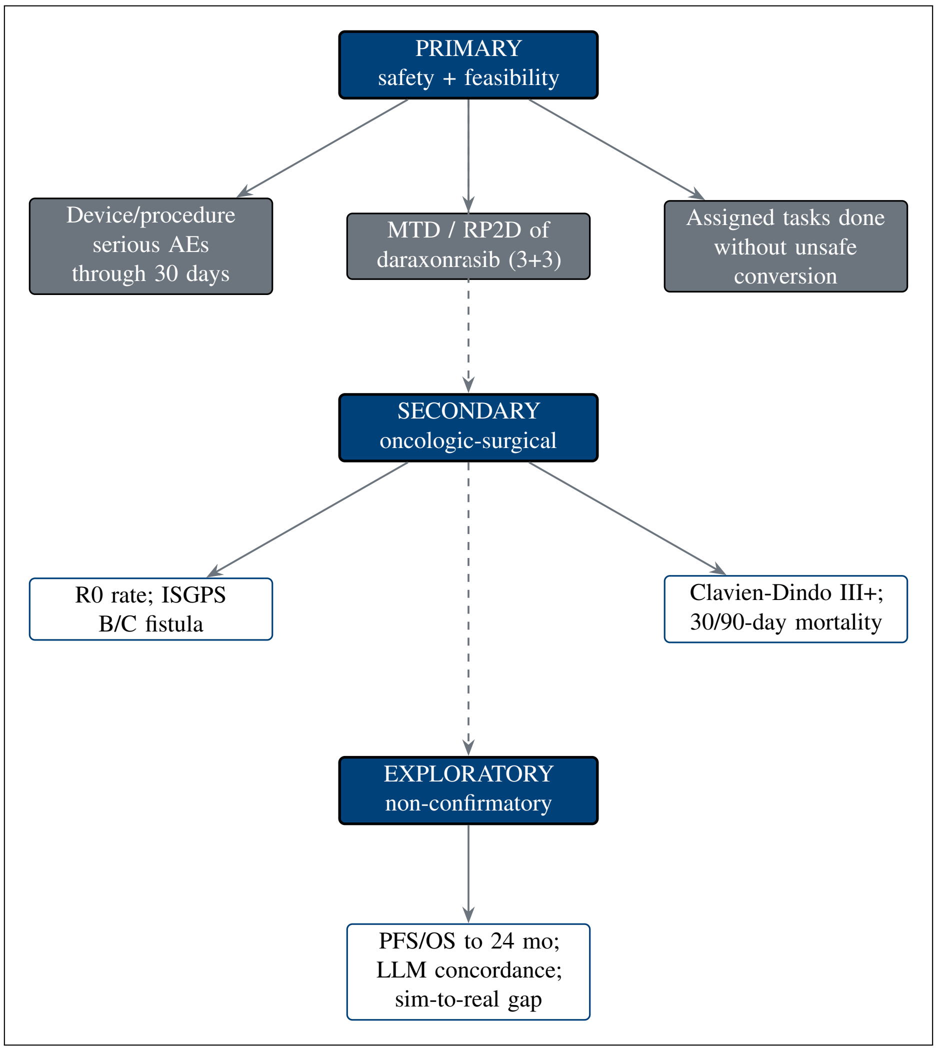

Figure 3. Objective-to-endpoint hierarchy. Primary safety and feasibility (30-day serious adverse event incidence, daraxonrasib MTD/RP2D, and unsafe-conversion-free task completion) sit above secondary oncologic-surgical quality (R0 rate, ISGPS grade B/C fistula, Clavien-Dindo grade III+, and 30- and 90-day mortality) and exploratory progression-free and overall survival, LLM advisory concordance, and the sim-to-real gap.

4.1 Primary Objectives and Endpoints

The study carries three co-primary objectives, one for each axis of the combined IND/IDE investigation: a drug-and-procedure safety objective, a drug dose-finding objective, and a device technical-feasibility objective.

The **primary safety objective** is to characterize the early safety of the combined intervention. Its endpoint is the incidence of device-related or procedure-related serious adverse events through 30 days after the procedure, graded for severity under the Clavien-Dindo classification, with grade III or higher events counted as the principal severe-complication signal. Attribution to the device or to the procedure is adjudicated by an independent reviewer who is blinded to the daraxonrasib dose level, so that drug-attributable toxicity and device-attributable or procedure-attributable harm are separated. The 30-day window is the standard first-in-human horizon for major hepatopancreatobiliary surgery and captures the dominant early lethal cascade of pancreatic fistula, intra-abdominal infection, hemorrhage, sepsis, and failure to rescue.

The **primary dose-finding objective** is to determine the MTD and RP2D of perioperative daraxonrasib. Its endpoint is defined operationally by the rate of dose-limiting toxicities within the 28-day DLT window of a standard 3+3 escalation across dose levels DL1 through DL3 (§4.3). The MTD is the highest dose level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD with integration of all safety, tolerability, and available pharmacokinetic data. Daraxonrasib eligibility and the dose anchor follow the broad pan-KRAS criteria of the RASolute 302 program [3].

The **primary technical-feasibility objective** is to establish that the eight-arm robotic Whipple can perform its intended function under continuous human oversight. Its endpoint is the proportion of procedures that complete all prespecified assigned operative tasks (dissection of the superior mesenteric and portal veins, hepatic-artery control, the three anastomoses, hemostasis, and drain placement) without an unplanned conversion caused by device malfunction or unsafe device behavior. A protocol-defined conversion undertaken for patient safety at the discretion of the supervising surgeon, in the absence of device malfunction, does not count against this endpoint but is recorded and reported. Task-completion without unsafe conversion is the early-feasibility readout recognized by the FDA for a significant-risk device [17].

4.2 Secondary Objectives and Endpoints

The secondary objectives characterize the oncologic adequacy and the surgical-quality profile of the combined intervention against the contemporaneous human benchmark. Each secondary endpoint is summarized descriptively with an exact binomial confidence interval; the study is not powered for formal hypothesis testing against the comparators.

The **R0 resection rate** is the proportion of participants with a microscopically margin-negative resection at day-90 pathology review, the single strongest modifiable surgical determinant of survival in resectable PDAC. The **ISGPS grade B/C postoperative pancreatic fistula rate** [4] is the leading driver of post-Whipple morbidity; the 2025 Dutch nationwide robotic cohort reported a grade B/C fistula rate of 24.4% [5]. The **Clavien-Dindo grade III or higher complication rate** through 90 days captures major complications requiring procedural, endoscopic, or intensive-care management; the Dutch comparator is 41.3% [5]. **Thirty-day and 90-day all-cause mortality** are the short-term survival safety measures, with a Dutch 30-day comparator of 3.9% [5]. The **unplanned conversion rate** from the robotic system to an open or conventional approach contextualizes the feasibility endpoint; the Dutch comparator is 10.1% [5]. Taken together, these comparators provide a transparent reference frame, but the small single-arm sample means they are interpreted as benchmarks rather than as targets for inferential comparison.

4.3 Exploratory Objectives and Endpoints

The exploratory objectives generate hypotheses for later-phase study and are explicitly non-confirmatory; no exploratory analysis is used to make a go or no-go decision on the intervention and all are reported with that caveat.

Progression-free survival and overall survival are assessed to 24 months from the day of surgery using Response Evaluation Criteria in Solid Tumors imaging at day 90 and every 12 weeks thereafter, providing a preliminary signal of oncologic benefit that the sample size cannot confirm. **LLM advisory concordance** is the proportion of LLM advisory outputs that agree with the independent hard-coded safety gate, quantifying the reliability of the second-opinion oracle and informing autonomy-graduation decisions reserved for later phases (§4.1). The **sim-to-real gap** is the difference between the Phase 0 simulation prediction and the intra-operative record, reported as the trajectory gap in millimeters (target less than 2 mm) and the tip-force gap in newtons (target less than 0.5 N); it validates the fidelity of the simulation-validation gate and underpins the digital-evidence basis for the device feasibility claim. Because these measures are descriptive, they are summarized with point estimates and dispersion only, without confirmatory significance testing.

5 Study Design

5.1 Overall Design

This is a prospective, open-label, single-arm, first-in-human study that combines a Phase 1 standard 3+3 dose-finding evaluation of perioperative daraxonrasib (RMC-6236) with an early feasibility evaluation of the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, a significant-risk Class II collaborative device. The study is conducted under a combined IND/IDE submission: the drug component proceeds under the investigational new drug pathway and the device component under the investigational device exemption pathway, with a single integrated protocol, a single informed consent that carries the Physical AI opt-out, and a single safety governance structure. The central hypothesis is that the combined intervention can be delivered with an acceptable early safety profile and can complete the assigned operative tasks under continuous human oversight while a recommended Phase 2 dose of daraxonrasib is identified, supporting progression to a later-phase efficacy study.

Enrollment is planned for up to $n = 18$ participants assigned across three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule with staggered (sentinel) enrollment, so that no two participants in a cohort undergo surgery within the same DLT-evaluation interval until the supervising surgeon and the data safety monitoring board (DSMB) have reviewed the preceding case. A mandatory Phase 0 simulation-validation gate precedes any patient-facing robotic activity (§4.3). The device operates within a phase-graduated staged-autonomy model: in this Phase 1 protocol the system runs only at the clinician-controlled (teleoperation or supervised) level and the supervised semiautonomous level, both under continuous human oversight with an immediate manual override; full autonomy is prohibited at this phase consistent with 21 CFR §312.21(e), and any future move to higher autonomy is reserved for a later phase and gated by independent safety review, a Unified Safety Level (USL) reassessment, and DSMB concurrence [18]. Figure 4 presents the staged-autonomy model.

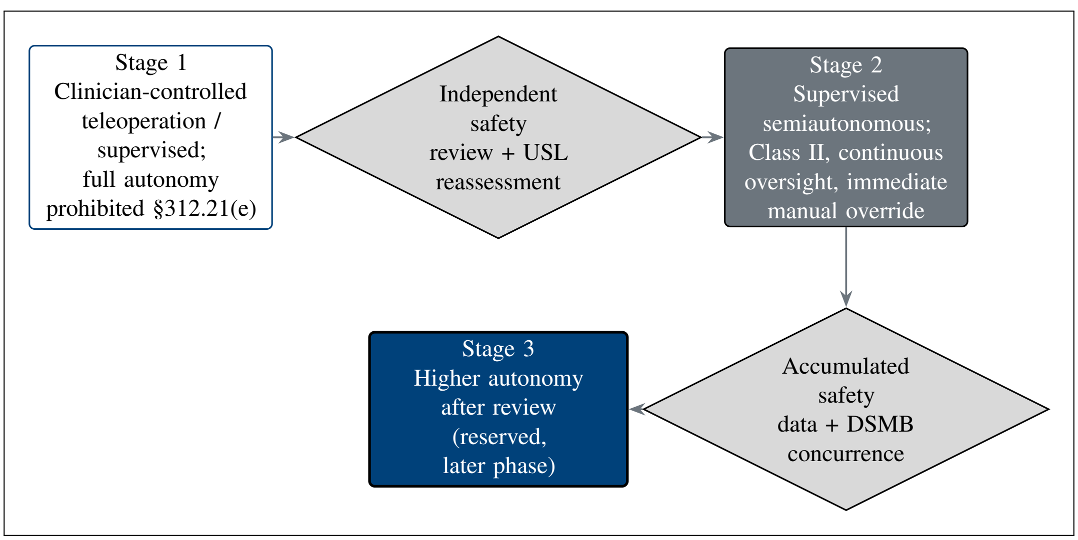

Figure 4. Phase-graduated staged-autonomy model. Autonomy increases only through gated review: this Phase 1 protocol operates at the clinician-controlled and supervised semiautonomous levels (Class II, continuous oversight with immediate manual override; full autonomy prohibited per 21 CFR §312.21(e)), and any move to higher autonomy requires an independent safety review, a USL reassessment, and DSMB concurrence, reserved for a later phase.

5.2 Scientific Rationale for Study Design

A single-arm, open-label design is scientifically and ethically appropriate for this first-in-human study, and a randomized comparator arm is not. The objectives at this stage are to characterize early safety, to identify a recommended Phase 2 dose, and to demonstrate that a novel surgical system can perform its intended function within hard cyber-physical limits; none of these is a comparative efficacy question, and each is answered most efficiently and most safely in a small, intensively monitored single-arm cohort. The early-feasibility pathway recognized by the FDA for significant-risk devices is explicitly structured around small first-in-human cohorts that generate the device-performance and safety evidence needed before any controlled comparison is undertaken [17]. Randomizing a vulnerable population to an unproven combined drug-device intervention before that feasibility and safety evidence exists would expose participants to risk without a commensurate scientific gain and would be difficult to justify under the equipoise and minimization-of-risk principles that govern human-subjects research.

The study population is a vulnerable one. Participants have a lethal malignancy with a total five-year survival below 13% [2], are facing the most technically demanding curative-intent abdominal operation, and may experience therapeutic misconception about an LLM-directed robotic system. The single-arm design concentrates oversight on each participant, permits sentinel staggered enrollment with case-by-case DSMB review, and allows immediate pausing of accrual on a prespecified safety signal, all of which protect participants more effectively than a parallel randomized structure would at this stage. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies provides a robust external human benchmark for the secondary surgical-quality endpoints (§3), so the absence of an internal control does not leave the surgical results uninterpretable [5]. For these reasons the design is a single-arm dose-finding and early-feasibility study, with a randomized controlled comparison deferred to a later phase once safety, the RP2D, and device feasibility are established.

5.3 Justification for Dose

Daraxonrasib dose and 3+3 escalation. Perioperative daraxonrasib is administered orally once daily. The escalation uses exact doses anchored to the RASolute 302 pan-KRAS program, in which the 300 mg once-daily dose is the established recommended Phase 2 dose [3]. To preserve a conservative first-in-human margin in the perioperative setting, the escalation begins below that reference dose and steps up to it: DL1 is 160 mg once daily, DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose. Three participants are enrolled at each level and observed over a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates to the next level. If one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level define that level as exceeding the MTD and fix the MTD at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD. Figure 5 shows the escalation scheme.

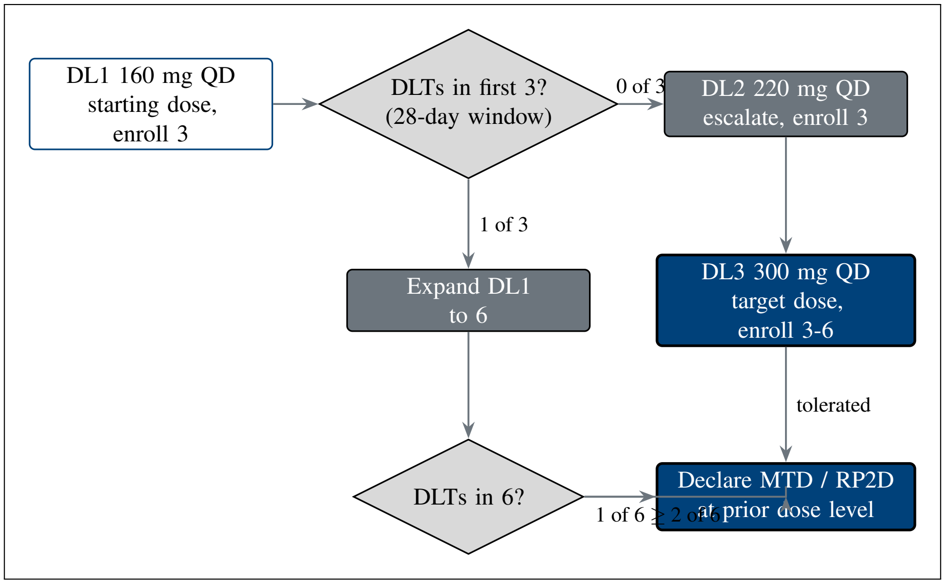

Figure 5. Daraxonrasib 3+3 dose-escalation scheme. Cohorts of three advance when no dose-limiting toxicity is observed in the 28-day window, expand to six on a single dose-limiting toxicity, and define the maximum tolerated dose and recommended Phase 2 dose at the highest tolerated level, escalating from DL1 (160 mg once daily) through DL2 (220 mg) to the DL3 reference dose (300 mg).

Device "dose" analogue and the Phase 0 gate. The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing activity, and it is escalated only by passing the mandatory Phase 0 simulation-validation gate. Three conditions must be met and documented before the first robotic case. First, the system must achieve a Unified Safety Level (USL) rating of at least 7.0 on the surgical-readiness scale. Second, the Phase 0 campaign must complete at least 1000 simulated procedures across at least two independent simulation frameworks, so that the readiness finding is not an artifact of any single simulator. Third, the sim-to-real fidelity must fall within prespecified limits: a trajectory gap of less than 2 mm and a tip-force gap of less than 0.5 N between the simulation prediction and the operative record [18]. These limits operate together with the intra-operative hard caps (a ≤ 3 ms cross-arm emergency-stop budget, per-arm tip-force caps of ≤ 3 N, a cumulative cross-arm cap of ≤ 18 N, and vascular no-fly gating around the superior mesenteric and portal veins), so that the device dose analogue is met only when both the simulation evidence and the real-time safety envelope are demonstrably in place.

5.4 End of Study Definition

An individual participant is considered to have completed the study at the last visit or procedure specified in the Schedule of Activities (§1.3), which is the 24-month overall-survival assessment, or at the participant's death, documented disease progression precluding further follow-up, or withdrawal of consent, whichever occurs first. The end of the study as a whole is defined as the date of the last scheduled assessment for the last participant, which is the last participant's 24-month overall-survival assessment. After that date, no further study-specific procedures are performed and the database is locked for the primary analysis. Participants who discontinue the investigational intervention for a protocol-defined reason (§7) remain in long-term follow-up for survival to the 24-month endpoint unless they withdraw consent for follow-up.

6 Study Population

The study population comprises adults with histologically confirmed, resectable or borderline-resectable, KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) who are eligible for peri-operative daraxonrasib and whose anatomy is compatible with the eight-arm robotic system. Target enrollment is approximately 36 participants screened to yield 18 treated, consistent with the sample size in §9. Figure 6 presents the CONSORT-style participant flow from screening through the 3+3 dose-escalation cohorts to the analysis populations.

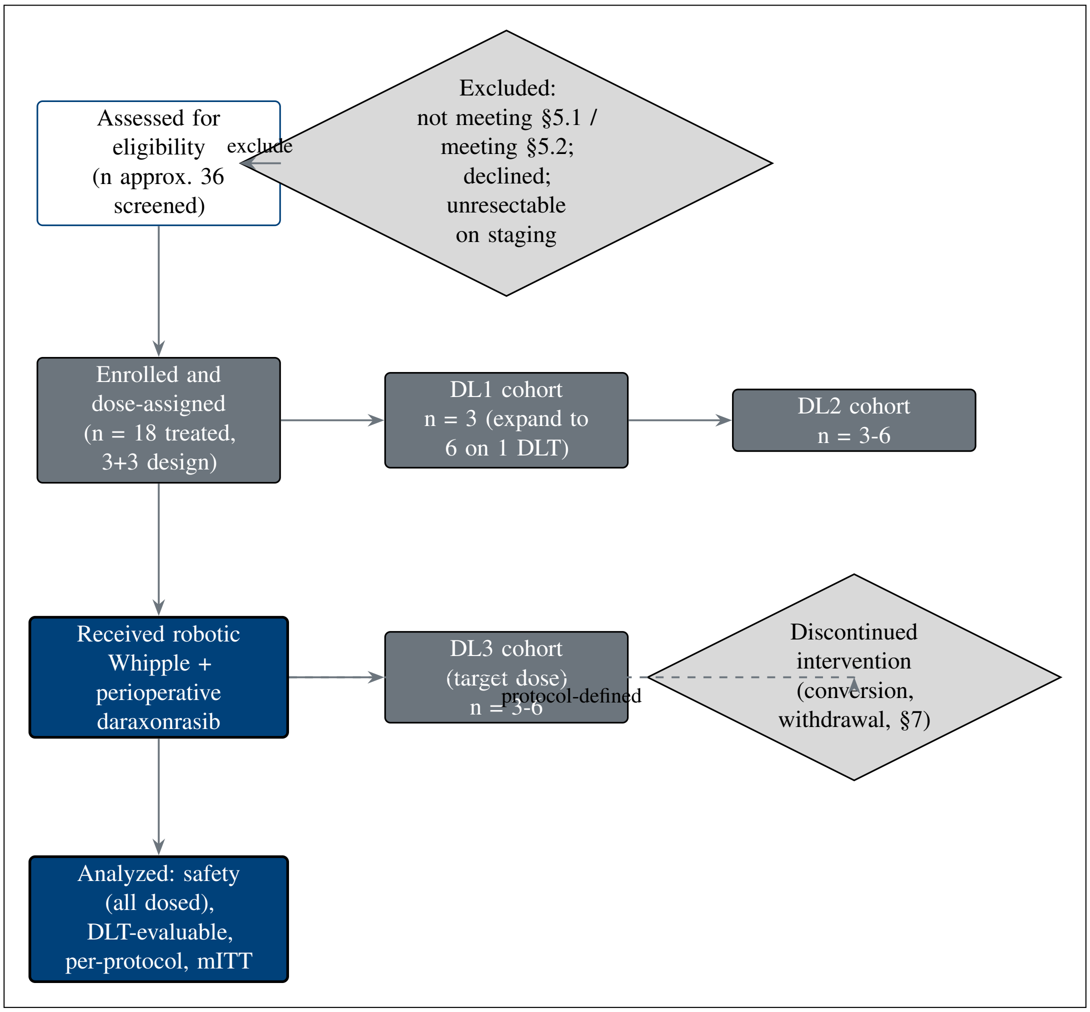

Figure 6. CONSORT-style participant flow from screening (approximately 36 assessed) through the 3+3 dose-escalation cohorts (DL1 through DL3, up to n = 18 treated) to the analysis populations, with the exclusion and protocol-defined discontinuation branches that feed the per-protocol and safety sets.

6.1 Inclusion Criteria

To be eligible to participate in this study, an individual must meet all of the following criteria.

1. Signed and dated informed consent obtained before any study-specific procedure, including the participant's explicit election or declination of the Physical AI opt-out, which governs use of the LLM-directed advisory layer (§8).

2. Age $\geq$ 18 years at the time of consent.
3. Histologically or cytologically confirmed PDAC that is resectable or borderline-resectable on multidisciplinary radiologic review.
4. A documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated tumor genotyping assay.
5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.
6. Adequate organ and marrow function on screening laboratories, including hematologic, hepatic, and renal indices sufficient to undergo major pancreaticobiliary surgery and to receive daraxonrasib.
7. A measured baseline CA 19-9 value (no threshold is imposed; the value is recorded for serial assessment).
8. Eligible for perioperative daraxonrasib under the broad pan-KRAS criteria of the RASolute 302 program [3].
9. Tumor and vascular anatomy compatible with the eight-arm robotic system on the preoperative imaging review, including port placement and working volume adequate for the assigned operative tasks.

The synthetic reference exemplar PAT-PDAC-0001 illustrates a representative eligible participant: a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75°, KRAS G12D, microsatellite stable, CA 19-9 of 412 U/mL at diagnosis, ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course of four cycles [6]. This exemplar is used for design illustration and does not represent an enrolled individual.

6.2 Exclusion Criteria

An individual who meets any one of the following criteria is excluded from participation.

1. Distant metastatic disease on staging.
2. Vascular involvement that precludes a margin-negative (R0) resection, or anatomy otherwise incompatible with the eight-arm robotic approach.
3. Prior pancreatic resection or prior surgery that precludes the planned reconstruction.
4. Uncontrolled intercurrent illness or comorbidity (for example uncontrolled cardiac, pulmonary, hepatic, renal, or active infectious disease) that would make major surgery or daraxonrasib administration unsafe.
5. Pregnancy or lactation, or unwillingness to use adequate contraception during the study where applicable.
6. Known contraindication or hypersensitivity to daraxonrasib or to a required excipient, or a concomitant medication interaction that cannot be safely managed.
7. Inability to provide informed consent or to comply with the Schedule of Activities.

Participant selection is equitable. Eligibility is determined solely by the scientific and safety criteria above, and no eligible individual is excluded on the basis of sex, gender, race, ethnicity, socioeconomic status, or insurance status. Limited English proficiency is not an exclusion criterion; qualified interpreters and translated consent materials are provided so that individuals with limited English proficiency can be fully informed and can participate on the same basis as English-speaking individuals.

6.3 Lifestyle Considerations

Perioperative lifestyle restrictions apply and are reviewed at consent and at the baseline visit. Participants follow a standardized perioperative nutrition plan, including the institutional preoperative fasting protocol and a staged postoperative diet advancement keyed to recovery of gastrointestinal function and to fistula status. Alcohol is to be avoided during the perioperative period and during any interval of active daraxonrasib dosing, both to limit hepatotoxic burden and to avoid confounding the safety assessment. Tobacco cessation is strongly advised and is supported with referral, given its effect on wound healing and pulmonary risk. No restriction on routine physical activity is imposed beyond standard postoperative surgical guidance. There are no study-specific dietary caffeine or grapefruit restrictions beyond those required by daraxonrasib labeling and the concomitant-medication rules in §6.

6.4 Screen Failures

A screen failure is an individual who consents to participate but is not subsequently enrolled because one or more eligibility criteria are not met. For each screen failure, a minimal CONSORT screen-failure data set is recorded: demography, the specific criterion or criteria that were not met (the screen-failure detail), the eligibility determination, and any serious adverse event that occurred between consent and the determination. No other study procedures are performed after a screen-failure determination. An individual who fails screening may be rescreened if the disqualifying condition was transient and is expected to resolve (for example a correctable laboratory abnormality); a rescreened individual retains the same participant number, and the rescreening is documented so that the screening-to-enrollment accounting in Figure 6 remains complete.

6.5 Strategies for Recruitment and Retention

Recruitment is conducted at a single academic medical center through the institutional multidisciplinary pancreatic-cancer program. Eligible individuals are identified at tumor-board review and at surgical and medical-oncology clinics, and the study is described to them by a member of the research team who is not the treating surgeon, to reduce the potential for undue influence. The planned accrual rate is approximately one to two participants per month, consistent with the sentinel staggered enrollment of the 3+3 design and the Phase 0 readiness gate, giving an enrollment period sufficient to treat 18 participants from approximately 36 screened.

Retention through the 24-month overall-survival window is supported by structured follow-up at day 30, day 90, and every 12 weeks thereafter, with appointment reminders, flexible scheduling, coordination with referring providers for participants who travel, and reimbursement of reasonable travel and parking costs. Because the population is vulnerable, additional safeguards consistent with Office for Human Research Protections (OHRP) guidance are applied: enrollment uses an independent consent monitor when indicated, the consent process explicitly addresses therapeutic misconception about the LLM-directed robotic system and the voluntariness of the Physical AI opt-out, and participants are reminded at each contact that they may withdraw at any time without affecting their clinical care. Compensation is limited to reimbursement of documented study-related travel, parking, and meal expenses, provided to the participant by the site after each completed study visit; no payment is offered for undergoing the surgical procedure or the investigational drug, so that compensation does not function as an undue inducement.

7 Study Intervention

The study intervention has two inseparable components delivered under a single integrated protocol: an investigational drug, perioperative daraxonrasib (RMC-6236), evaluated by a standard 3+3 dose-finding design under the investigational new drug pathway; and an investigational device, the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, evaluated as an early-feasibility significant-risk Class II collaborative platform under the investigational device exemption pathway. This section describes both components, their administration, their preparation and accountability, the measures taken to minimize bias, the compliance machinery for each, and the permitted and prohibited concomitant therapy. The two components are coupled at one decision point only: the perioperative pause-and-restart advisory for the drug is informed by the operative and pathology results of the device procedure (§6.1.2).

7.1 Study Intervention Administration

7.1.1 Study Intervention Description

Drug component: daraxonrasib (RMC-6236). Daraxonrasib is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the active, guanosine triphosphate bound state of mutant RAS in complex with cyclophilin A, suppressing downstream effector signaling across the common KRAS oncoprotein variants found in pancreatic ductal adenocarcinoma. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic pancreatic ductal adenocarcinoma [8], and its dose and exposure assumptions in this protocol are anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established recommended Phase 2 dose [3]. In this study daraxonrasib is administered orally, once daily, in a perioperative schedule: a defined pre-operative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by the advisory described in §6.1.2. The drug is investigational in this perioperative, curative-intent surgical context and for the combined drug-device use described here.

Device component: on-premises LLM-directed eight-arm Whipple system. The investigational device is an eight-arm parallel coelomic surgical platform in which an on-premises repository-pinned LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack, never as an autonomous controller. The platform carries 56 mechanical degrees of freedom (8×7 per-arm degrees of freedom) and a 640-channel mixed-rate sensor stack (80 channels per arm). Force channels sample at 100 kHz; all remaining channels sample at the 10 kHz command rate. Positioning accuracy is 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. The hard cyber-physical safety envelope comprises a per-arm tip-force cap of ≤ 3 N, a cumulative cross-arm tip-force cap of ≤ 18 N enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a $100\,\mu$s per-arm watchdog deadline, an emergency arm park within $50\,\mu$s of a missed or out-of-parameter frame, and a cross-arm emergency stop (E-stop) that halts the full platform in ≤ 3 ms. A software-independent hardware E-stop backs the software chain. In this Phase 1 protocol the device operates only at the clinician-controlled and supervised semiautonomous autonomy levels under continuous human oversight with immediate manual override; full autonomy is prohibited consistent with 21 CFR §312.21(e). The advisory control loop and the platform architecture are shown in Figure 6 and Figure 7.

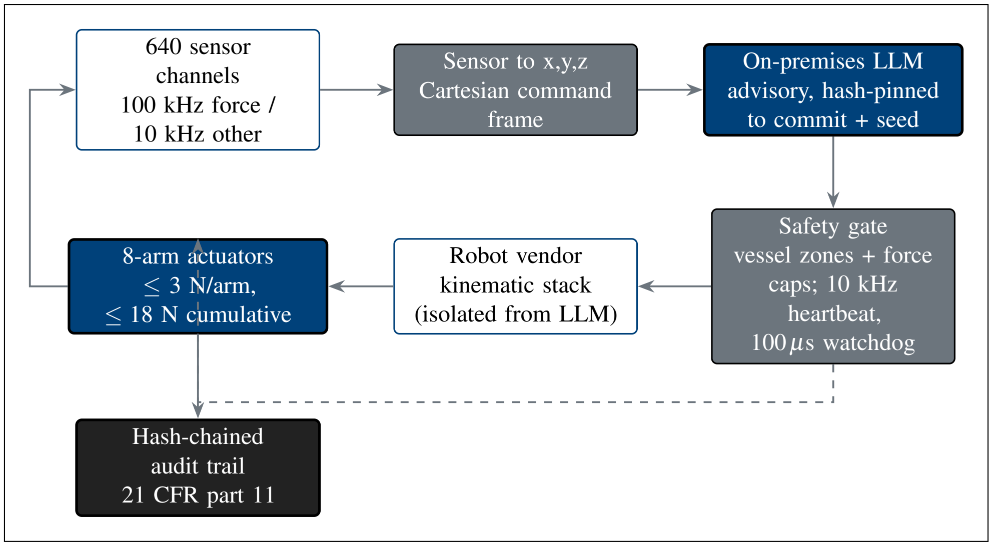

Figure 6. On-premises LLM advisory control loop. Sensor data is mapped to a Cartesian command frame; the LLM issues hash-pinned advisory commands from outside the vendor kinematic stack (second-opinion oracle isolation); and a safety gate enforces the vessel zones, the force caps, the 10 kHz heartbeat, the $100\,\mu s$ *watchdog, and the* ≤ 3 *ms E-stop before any actuator moves. The actuator telemetry returns to the sensor stack and every step is written to a 21 CFR part 11 hash-chained audit trail.*

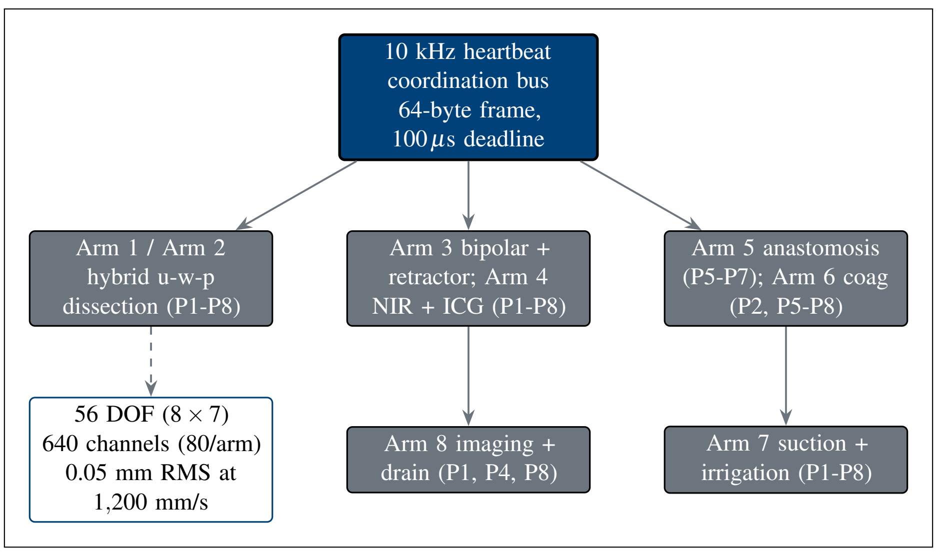

Figure 7. Eight-arm platform architecture. Eight cooperating arms (56 degrees of freedom, 640 sensor channels at 80 per arm, 0.05 mm root mean square positioning at 1,200 *mm/s) are coordinated by a 10 kHz heartbeat bus, with each arm's tool assignment and operative-phase coverage driving the Schedule of Activities.*

The per-arm tool assignment crossed with the eight operative phases (P1 through P8) is given in Table 6, and ten representative channels drawn from the 640-channel inventory are given in Table 7.

Physical AI System Description fields (21 CFR §312.23(g)). Because the device participates in the investigational drug delivery pathway, the combined IND/IDE carries a full Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g) [1]. The fields, and the location of the corresponding evidence in this submission, are summarized below.

Table 1: Per-arm tool assignment and operative-phase coverage for the eight-arm LLM-directed Whipple platform.

Arm	Primary tool	Phase coverage	Sensor channels (10 of 80)
Arm 1	Hybrid u-w-p end-effector	P1 to P8 (active dissect)	Tip force, joint torque, ID
Arm 2	Hybrid u-w-p end-effector	P1 to P8 (active dissect)	Tip force, joint torque, ID
Arm 3	Bipolar + retractor	P1 to P8 (vessel control)	Tip force, current, force
Arm 4	NIR + ICG imaging probe	P1 to P8 (imaging)	ICG fluorescence, NIR depth
Arm 5	Anastomosis probe	P5, P6, P7 (idle others)	Ring tension, ring strain
Arm 6	Bipolar coag + cautery	P2, P5, P6, P7, P8	Tip force, RF power
Arm 7	Suction + irrigation	P1 to P8	Suction pressure, pH
Arm 8	Final imaging + drain placement	P1, P4, P8 (light load)	ICG fluorescence, drain ID

Table 2: Ten representative sensor channels drawn from the 640-channel inventory (80 channels per arm).

Arm	Channel name	Rate	Unit	Dynamic range
1	tip_force_x	100 kHz	N	0 to 3.0 N
1	joint_torque_3	10 kHz	N·m	−50 to 50
2	tip_force_z	100 kHz	N	0 to 3.0 N
3	retractor_force	10 kHz	N	0 to 5.0 N
4	icg_fluorescence	10 kHz	a.u.	0 to 65,535
4	nir_depth	10 kHz	mm	0 to 50
5	ring_tension	10 kHz	N	0 to 1.0 N
6	rf_power	10 kHz	W	0 to 80
7	suction_pressure	10 kHz	kPa	−90 to 0
8	drain_id	10 kHz	enum	0 to 4

(i) *System identification and classification:* manufacturer, model designation, software version, surgical robot category, and the current Unified Safety Level (USL) score computed across simulation switching, AI integration, cross-robot sharing, and clinical-trial collaboration.

(ii) *Role in drug administration:* secondary role (the device does not itself dispense daraxonrasib; it conducts the resection and reconstruction and generates the operative and pathology inputs to the restart advisory), with the permitted autonomy levels and human-oversight requirements for each procedure.

(iii) *Hardware specifications:* 56 degrees of freedom, 0.05 mm root mean square positioning at 1,200 mm/s, 100 kHz force sensing, the ≤ 3 N per-arm and ≤ 18 N cumulative tip-force limits, and end-effector details.

(iv) *Software and AI/ML specifications:* the operating and middleware stack, the control algorithms, and the on-premises advisory LLM together with its training data characteristics, validation methodology, performance metrics, and Predetermined Change Control Plan (PCCP).

(v) *Simulation validation data:* the Phase 0 campaign results, including cross-framework consistency and the sim-to-real gap quantification (trajectory gap < 2 mm, tip-force gap < 0.5 N).

(vi) *Digital twin specifications:* the digital-twin methodology, calibration sources, update frequency, and validation metrics used for operative rehearsal and intra-operative guidance.

(vii) *Safety systems:* the emergency-stop mechanisms (hardware and software), force and torque limits, collision detection, workspace-boundary enforcement, fail-safe behaviors, and the pre-procedure safety matrix.

(viii) *Cybersecurity assessment:* the network architecture, deny-by-default authentication and authorization, encryption, vulnerability management, and incident response.

(ix) *MCP integration:* the Model Context Protocol server topology, conformance level, robot capability profile, and the hash-chained audit-trail implementation under 21 CFR part 11 [19].

(x) *Human-robot interaction protocols:* the operator control stations, status displays, alarm systems, handoff procedures, and the training requirements for operating and supervising personnel.

(xi) *Maintenance and calibration:* the scheduled maintenance intervals, calibration procedures, out-of-service criteria, and re-qualification requirements (§6.2).

7.1.2 Dosing and Administration

Daraxonrasib dosing and the 3+3 escalation. Daraxonrasib is administered orally once daily at one of three dose levels assigned by the 3+3 rule with sentinel staggered enrollment. The escalation begins below the RASolute 302 reference dose to preserve a conservative first-in-human margin and steps up to it: DL1 is 160 mg once daily, DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose [3]. Three participants are enrolled at each level and observed across a 28-day dose-limiting toxicity (DLT) window. If no DLT occurs among the first three, the cohort escalates. If one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level fix the maximum tolerated dose (MTD) at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the recommended Phase 2 dose is selected at or below the MTD. The escalation scheme is detailed in §4.3.

Perioperative pause-and-restart advisory. Daraxonrasib is paused before surgery and restarted postoperatively on a day chosen by an advisory that is LLM-bound and human-confirmed, never autonomous. The advisory takes two inputs: the pancreaticojejunostomy ISGPS postoperative pancreatic-fistula grade (A, B, or C) [4] and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. It recommends a restart on postoperative day T+7d, T+14d, or T+21d (or a continued hold). In the 32-iteration deterministic simulation sweep, baseline serum was 0.45 ng/mL at the operative timepoint across all iterations (below the 0.5 ng/mL trough), and the advisory recommended T+7d in 29 of 32 iterations (Grade A fistula with sub-threshold trough), T+14d in 3 of 32 (Grade B or delayed serum rise), and T+21d in 0 of 32. Grade C fistula maps to T+21d or a continued hold and to the drug stop rule of §7.1. The advisory logic is shown in Figure 8.

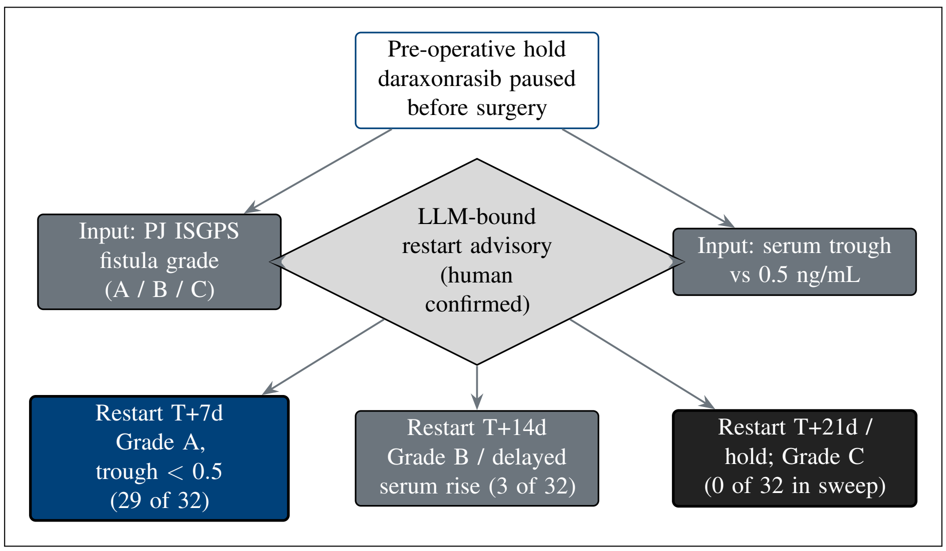

Figure 8. Daraxonrasib perioperative pause-and-restart advisory. The restart day is keyed to the pancreaticojejunostomy ISGPS fistula grade and the serum trough relative to 0.5 ng/mL: T+7d for Grade A with a sub-threshold trough (29 of 32 simulation iterations), T+14d for Grade B or a delayed serum rise (3 of 32), and T+21d or continued hold for Grade C (0 of 32 in the sweep). The advisory is bound and human-confirmed, never autonomous.

Device "administration": operative phases and the three anastomoses. The device has no pharmacologic dose; its administration is the conduct of the pancreaticoduodenectomy across eight operative phases (P1 through P8) under continuous human oversight. Phases P1 through P4 comprise the dissection and specimen removal (Arms 1 and 2 dissecting under Arm 3 vessel control and Arm 4 near-infrared indocyanine-green imaging); the specimen is removed at the end of P4 and reconstruction begins. The three reconstructive anastomoses are then constructed in sequence under closed-loop ring-tension control by the Arm 5 anastomosis probe, which issues a soft warning whenever measured ring tension falls outside the controller band: the duct-to-mucosa pancreaticojejunostomy (PJ) in P5 with a band of 0.30 to 0.60 N, the end-to-side hepaticojejunostomy (HJ) in P6 with a band of 0.20 to 0.50 N, and the antecolic gastrojejunostomy (GJ) in P7 with a band of 0.40 to 0.80 N; P8 covers final imaging, hemostasis, and drain placement. The pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk and is therefore the anastomosis on which the fistula-grade input to the drug advisory is based. The reconstruction sequence and the ring-tension bands are shown in Figure 9.

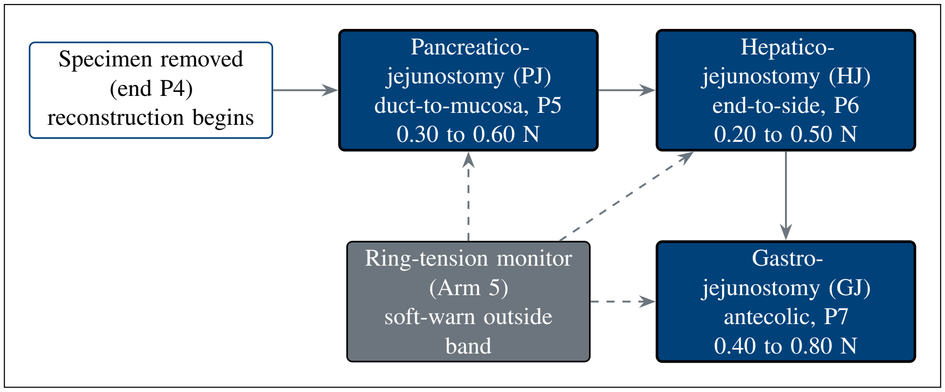

Figure 9. The three reconstructive anastomoses and their controller ring-tension bands: the duct-to-mucosa pancreaticojejunostomy (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), each guarded by the Arm 5 ring-tension monitor that issues a soft warning outside the band.

7.2 Preparation, Handling, Storage, and Accountability

7.2.1 Acquisition and Accountability

Daraxonrasib is supplied by the sponsor to the investigational-product pharmacy at each participating site and is dispensed only by the site pharmacist or an authorized designee against a written prescription tied to an enrolled, eligible participant and a confirmed dose level. The pharmacy maintains a drug-accountability log recording, for every unit, the lot number, the quantity received, the quantity dispensed, the quantity returned, and the quantity destroyed, reconciled against shipment records at each monitoring visit. Returned and unused product is retained for reconciliation and destroyed only after sponsor authorization. The investigational device is provided under the IDE and is not a consumable; its accountability is established through the asset record, the per-procedure pre-procedure safety matrix, and the hash-chained audit trail that captures every operative use (§6.4). Custody of both components is restricted to authorized, trained personnel, and all transfers are documented.

7.2.2 Formulation, Appearance, Packaging, and Labeling

Daraxonrasib is provided as oral tablets in the strengths required to compose the DL1 (160 mg), DL2 (220 mg), and DL3 (300 mg) once-daily doses. Tablets are packaged in child-resistant, light-protective containers labeled in accordance with 21 CFR §312.6, bearing the caution statement "Caution: New Drug. Limited by Federal (or United States) law to investigational use," together with the protocol number, lot number, storage conditions, and a unique container identifier; no information that would unblind an outcome assessor appears on the label. Because this is an open-label study, the participant-facing label identifies the product and the assigned daily dose. The investigational device and its single-use accessories are labeled per 21 CFR part 812 with the investigational-device caution statement and are not relabeled or repackaged at the site.

7.2.3 Product Storage and Stability

Daraxonrasib is stored at controlled room temperature in a secured, access-controlled, temperature-monitored location within the investigational-product pharmacy, in its original light-protective packaging, until dispensed. Continuous temperature monitoring with documented excursion handling is maintained; any excursion outside the labeled range is quarantined and is not dispensed until the sponsor confirms continued suitability against the stability data on file. Stability is supported by the sponsor's stability program, and product is dispensed only within its labeled expiry or retest date. The investigational device is stored and maintained in its qualified clinical environment between procedures under the maintenance and calibration program below.

7.2.4 Preparation and Device Maintenance and Calibration

Drug preparation. No compounding or reconstitution is required; preparation consists of the pharmacist verifying the participant identity, the assigned dose level, and the expiry, then dispensing the once-daily oral dose with administration instructions. Tablets are swallowed whole with water and are not crushed, split, or chewed unless a future protocol amendment provides validated instructions.

Device maintenance and calibration (21 CFR §312.23(g)(xi)). Before each clinical procedure the system must pass the pre-procedure safety matrix: verification of the 640-channel sensor stack, confirmation of the per-arm force calibration against the ≤ 3 N per-arm and ≤ 18 N cumulative caps, exercise of the 10 kHz heartbeat with its $100\,\mu$s watchdog, a successful software and hardware E-stop self-test, and confirmation that the positioning accuracy remains within the 0.05 mm root mean square specification at 1,200 mm/s. Scheduled maintenance, periodic force and positional calibration, and performance verification are performed at the manufacturer-specified intervals and documented. A system that fails any pre-procedure check, exhibits an out-of-tolerance calibration, or experiences a hard-stop E-stop event is taken out of service and is returned to clinical use only after re-qualification against the same matrix and sponsor authorization. All maintenance, calibration, and out-of-service events are recorded in the hash-chained audit trail.

7.3 Measures to Minimize Bias: Randomization and Blinding

This is an open-label, single-arm, first-in-human study; there is no randomization and no comparator arm, consistent with the dose-finding and early-feasibility objectives (§4.2) and with ICH E9 principles for early-phase trials. Because no treatment assignment is masked, bias control is concentrated on the outcome assessments most vulnerable to it. The primary surgical-efficacy endpoint, R0 resection status, is determined by a pathologist who is masked to the assigned daraxonrasib dose level, to the intra-operative device telemetry, and to the LLM advisory record, and who applies a standardized synoptic margin protocol; discordant or indeterminate margins are adjudicated by an independent masked second pathologist. Radiographic response is read centrally against RECIST by a reviewer blinded to dose level and to clinical course, and the serious adverse-event and Physical AI safety adjudications described in §8.3 are performed by reviewers independent of the operative team. Sentinel staggered enrollment with case-by-case data safety monitoring board review further protects the integrity of the safety read-out by preventing within-cohort contamination of clinical judgment.

7.4 Study Intervention Compliance

Daraxonrasib adherence. Adherence is documented through the dispensing and accountability log, returned-tablet counts at each visit, and a participant dosing diary reconciled against the pharmacy record. Adherence is corroborated objectively by the serum daraxonrasib assay, which both confirms exposure and supplies the trough value used by the restart advisory (§6.1.2); a documented trough discordant with the reported intake prompts adherence counseling and is recorded in the case report form. Material non-adherence is captured as a protocol deviation and informs the DLT-evaluability determination (§9).

Device compliance. Device compliance is enforced before and during every procedure rather than reconstructed afterward. No operative phase may begin until the pre-procedure safety matrix passes in full, and the 10 kHz heartbeat, the 100 μs watchdog, the force caps, and the vascular safety-zone gate operate continuously throughout the case. Every advisory command, every safety-gate verdict, every force and trajectory sample at the recorded rates, and every E-stop event is written to a 21 CFR part 11 hash-chained, append-only audit trail [19], so that compliance with the cyber-physical envelope is verifiable for each case and any deviation is detectable and time-stamped. The audit trail also satisfies the -24 h to $+72$ h preservation requirement that applies around a reportable Physical AI adverse event (§8.3.6).

7.5 Concomitant Therapy

All medications, biologics, blood products, radiotherapy, and clinically significant non-pharmacologic interventions taken from the time of informed consent through the end of safety follow-up are recorded in the case report form with indication, dose, route, and dates. Standard perioperative supportive care is permitted and encouraged, including antiemetics, analgesia, venous thromboembolism prophylaxis, peri-operative antimicrobial prophylaxis, proton-pump inhibitors, and pancreatic-enzyme replacement as clinically indicated, with attention to agents that may interact with daraxonrasib exposure. Prohibited during study participation are other systemic anticancer therapy (cytotoxic chemotherapy, other targeted agents, or immunotherapy) outside the protocol, other investigational agents or investigational devices, and any concomitant use of a second surgical robotic system for the index operation. Strong inducers or inhibitors of the principal metabolic pathway of daraxonrasib are avoided where feasible; when an interacting agent is clinically unavoidable, its use is documented and reviewed against the exposure data.

7.5.1 Rescue Medicine

Rescue is available along two axes. For surgical and device-related complications, the operative team may at any time convert to manual or open technique (§7.1) and deliver standard rescue care: transfusion and hemostatic resuscitation for bleeding, vasoactive support for hemodynamic instability, percutaneous or operative drainage and broad-spectrum antimicrobials for a clinically relevant postoperative pancreatic fistula or intra-abdominal collection, and somatostatin-analogue therapy where indicated for high-output fistula. For drug toxicity, daraxonrasib is held or discontinued per the DLT and stopping rules (§7.1) and managed symptomatically with the appropriate supportive agents (for example antidiarrheals and aggressive dermatologic care for the rash and gastrointestinal toxicities characteristic of the class), with dose interruption and reduction governed by the protocol. All rescue interventions are recorded as concomitant therapy and, where the criteria are met, as adverse or serious adverse events.

8 Intervention and Participant Discontinuation/Withdrawal

This section defines the prospectively specified rules for discontinuing the study intervention (the device, the drug, or both), the procedures governing a participant's withdrawal from the study, and the handling of participants lost to follow-up. Discontinuation of an intervention is distinct from withdrawal from the study: a participant who stops one or both components for a protocol-defined reason remains in the study for safety and survival follow-up unless consent for that follow-up is withdrawn.

8.1 Discontinuation of Study Intervention

Device stopping rules. The investigational device is discontinued for the index procedure, with conversion to the appropriate fallback technique, whenever any of the following prespecified conditions occurs. These rules implement the Physical AI stopping criteria of 21 CFR §312.23(g)(i)(v) and the divergence-reporting threshold of §312.32(g)(v) [1].

- *Conversion to open or manual technique.* Any intra-operative judgment by the supervising surgeon that patient safety or oncologic adequacy is better served by manual robotic or open conversion (including uncontrolled bleeding, unfavorable anatomy, or loss of a critical view) ends device-directed operation immediately; conversion is always available and is never itself a protocol violation.
- *Repeated force-limit excursions.* Repeated excursions against the ≤ 3 N per-arm or ≤ 18 N cumulative cross-arm tip-force caps, or any sustained or consecutive pattern of force-clamp activations indicating that the case cannot proceed within the force envelope, trigger discontinuation.
- *No-fly-zone violations.* Any breach of a vascular no-fly zone around a named vessel (superior mesenteric vein or portal vein at 1.0 mm; hepatic artery, celiac axis, or superior mesenteric artery at 1.5 mm), or a recurring pattern of no-fly approaches, ends device-directed dissection in that field.
- *Sim-to-real divergence beyond twice tolerance.* Any instance in which the observed device behavior diverges from the validated simulation prediction by more than twice the sim-to-real gap tolerance (a trajectory deviation greater than 4 mm or a force discrepancy greater than 1.0 N) triggers discontinuation and a report under §8.3.6.
- *Hard-stop E-stop event.* Any hard-stop cross-arm E-stop (the ≤ 3 ms full-platform halt, whether software- or hardware-initiated) ends the device-directed segment; resumption with the device requires that the cause be identified and that the system re-pass the pre-procedure safety matrix (§6.2), failing which the case is completed by conversion.

Following any device discontinuation, the operation is completed by the most appropriate conventional means, the hash-chained audit trail is preserved across the event window (§8.3.6), and the event is reviewed by the Physical AI Safety Review Committee and the data safety monitoring board before the device is used in a subsequent case.

Drug stopping rules. Daraxonrasib is discontinued in an individual participant for a dose-limiting toxicity attributable to the drug during the 28-day DLT window, for any other clinically significant or unacceptable toxicity that does not resolve with protocol dose interruption and reduction, for a confirmed ISGPS Grade C postoperative pancreatic fistula [4] (which precludes restart and maps to the T+21d or continued-hold arm of the perioperative advisory, §6.1.2), for pregnancy, for intercurrent illness that contraindicates continuation, or at the participant's request. Dose interruptions and reductions for lower-grade toxicity follow the protocol algorithm and do not by themselves constitute permanent discontinuation; a participant who permanently discontinues the drug remains in the study for safety and survival follow-up.

8.2 Participant Discontinuation/Withdrawal from the Study

A participant may withdraw from the study at any time, for any reason, without penalty and without loss of any benefit to which the participant is otherwise entitled, and a participant may separately withdraw consent for continued study-specific procedures while permitting ongoing survival follow-up through the medical record. The investigator may withdraw a participant for safety, for intercurrent illness, for significant non-compliance that compromises participant safety or data integrity, for loss of eligibility, or at the sponsor's direction if the study is terminated. The reason for and date of any discontinuation or withdrawal are recorded in the case report form, and a participant who withdraws is asked to complete a final safety assessment where feasible.

Data and specimens collected up to the point of withdrawal are retained and analyzed in accordance with the consent and applicable regulation; withdrawal of consent for future participation does not require deletion of data already collected. The relationship of discontinuation and withdrawal to the analysis populations (the safety, DLT-evaluable, and modified intention-to-treat populations) is defined in the statistical considerations (§9). A participant who withdraws before completing the operative intervention is replaced for the purpose of completing the 3+3 cohort only if not DLT-evaluable, per the rules in §9.

8.3 Lost to Follow-Up

A participant is considered lost to follow-up only after the site has made and documented at least three contact attempts by telephone across separate days, followed by a written contact attempt to the last known address, over the 24-month overall-survival window. Each attempt is recorded with its date and outcome in the case report form. Before declaring a participant lost to follow-up, the site attempts, with appropriate authorization, to ascertain vital status from the medical record, the referring provider, and permissible public sources, because survival is a key endpoint of the study. A participant who cannot be reached after these attempts is classified as lost to follow-up, with the last date known alive recorded. The handling of the resulting missing data, including censoring rules for progression-free and overall survival and the sensitivity analyses used to assess the impact of missingness, is specified in §9.4.

9 Study Assessments and Procedures

This section specifies the efficacy and safety assessments, the device-telemetry measurements unique to the Physical AI component, and the adverse-event and unanticipated-problem machinery for the study. The timing of every assessment is governed by the Schedule of Activities (§1.3); this section defines what is measured, how it is graded, and how it is reported.

9.1 Efficacy Assessments

The primary surgical-efficacy endpoint is the rate of margin-negative (R0) resection, determined by histopathology under a standardized synoptic protocol and read by a pathologist masked to the assigned daraxonrasib dose level and to the device telemetry (§6.3), with masked independent adjudication of discordant or indeterminate margins. Radiographic tumor response is assessed against RECIST version 1.1 on contrast-enhanced cross-sectional imaging read centrally by a blinded reviewer at the protocol-specified timepoints. Progression-free survival and overall survival are followed to the 24-month endpoint, with overall survival benchmarked descriptively against the 2025 Dutch nationwide robotic pancreaticoduodenectomy cohort.

Anastomosis quality is a prespecified feasibility endpoint and is assessed for each of the three reconstructions against its controller ring-tension band: the duct-to-mucosa pancreaticojejunostomy (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), as constructed under Arm 5 monitoring (§6.1.2). Postoperative pancreatic fistula is graded by the International Study Group of Pancreatic Surgery (ISGPS) definition as a biochemical leak (no longer classified as a clinically relevant fistula), Grade B (requiring a change in management), or Grade C (requiring reoperation or causing organ failure or death) [4]; the pancreaticojejunostomy grade is the dominant fistula determinant and is the input to the perioperative drug-restart advisory (§6.1.2) and to the drug stopping rule (§7.1).

9.2 Safety and Other Assessments

Clinical safety assessments comprise vital signs, a directed physical examination, hematology, serum chemistry including hepatic and pancreatic indices, coagulation studies, the serum daraxonrasib trough assay, and protocol-specified imaging, at the frequencies in the Schedule of Activities. In addition to these conventional assessments, the device generates a continuous safety telemetry stream that is itself a source of protocol observations under 21 CFR §312.23(g)(i)(iv): per-arm and cumulative force profiles, trajectory accuracy against the 0.05 mm root mean square specification, every vascular safety-zone verdict, every heartbeat and watchdog event, and every E-stop activation, each captured at its recorded rate in the hash-chained audit trail. The pre-procedure safety matrix result is recorded for every case.

The vascular safety-zone gate ticks at 10 kHz and evaluates the instrument-tip proximity to each of five named vessels against three concentric thresholds, with the verdict driving the response: a soft-warning zone at 3.0 mm raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone at 5.0 mm forces a ≤ 3 ms E-stop. Across the representative 100-tick verdict sample the gate returned clear on 83 ticks, soft_warning on 6, no_fly on 6, and hard_stop on 5, exercising all five vessels across all eight phases (40 vessel-by-phase cells). The gate is shown in Figure 13.

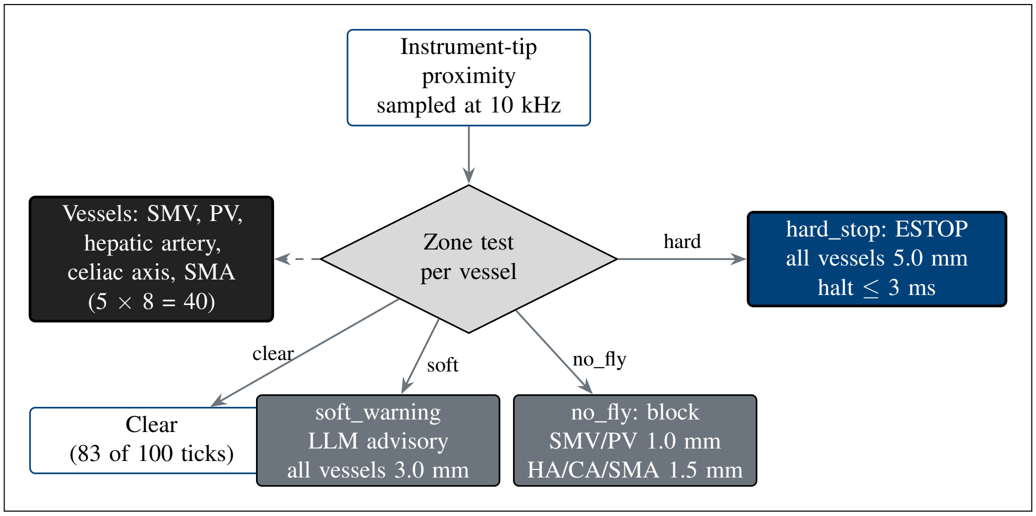

Figure 13. Five-vessel vascular safety-zone gate. Three concentric proximity zones around each named vessel drive the response: a soft-warning zone (3.0 mm) raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone (5.0 mm) forces a ≤ 3 ms E-stop. The gate is sampled at 10 kHz across all eight phases.

The halt chain that backs the gate is layered. The 10 kHz heartbeat bus broadcasts a 64-byte frame on a 100 μs per-arm watchdog deadline; a frame that is on time continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 μs and escalates to a cross-arm E-stop that halts the platform in $\leq$ 3 ms at a 0.0 N force clamp. A software-independent hardware E-stop meeting the 21 CFR §312.404 $\leq$ 500 ms requirement backs the software chain. The architecture is shown in Figure 14.

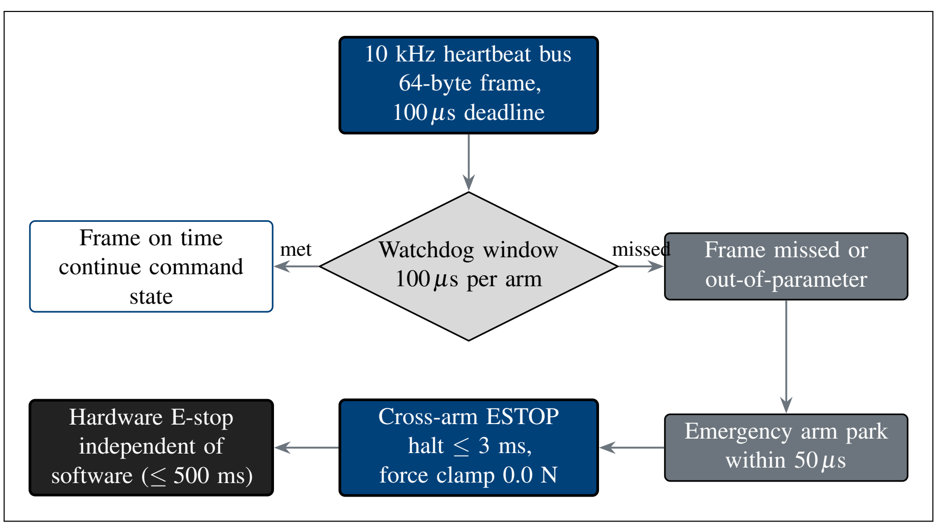

Figure 14. Heartbeat, watchdog, and E-stop architecture. A 10 kHz heartbeat bus with a 100 μs per-arm watchdog continues the command state on an on-time frame; a missed frame parks the arm within 50 μs and escalates to a cross-arm E-stop that halts the platform in $\leq$ 3 ms at a 0.0 N force clamp, backed by a software-independent hardware E-stop meeting the §312.404 $\leq$ 500 ms requirement.

9.3 Adverse Events and Serious Adverse Events

Adverse events are captured through two parallel and equally binding streams: a conventional pharmacovigilance stream for clinical events, and a Physical AI safety stream for device- and AI-related events. The combined workflow is shown in Figure 15.

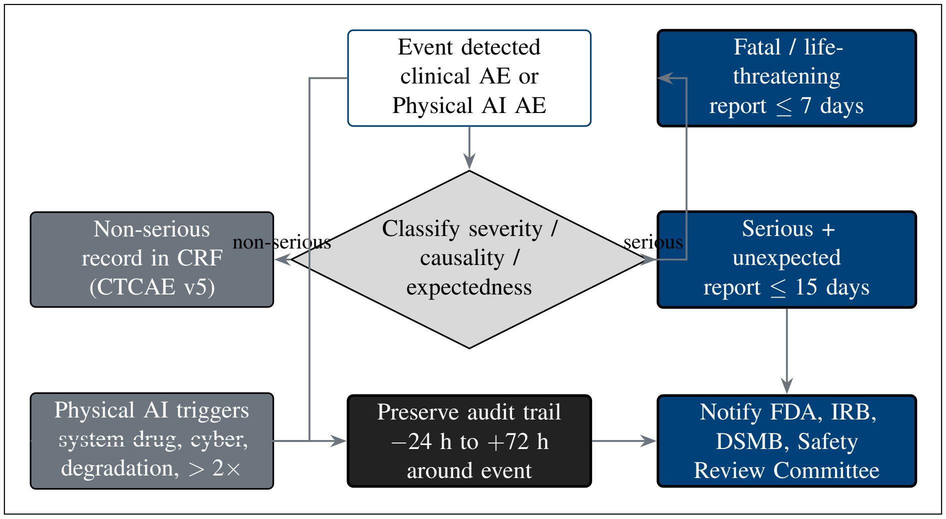

Figure 15. Adverse-event and Physical AI AE reporting workflow. Clinical events are graded by CTCAE v5 and reported on the 15-day (serious, unexpected) or 7-day (fatal, life-threatening) IND/IDE timelines, while Physical AI triggers add their own reports, each preserving the hash-chained audit trail from 24 hours before to 72 hours after the event, with notification to the FDA, the IRB, the DSMB, and the Physical AI Safety Review Committee.

9.3.1 Definition of Adverse Events (AE)

An adverse event is any untoward medical occurrence in a participant administered the study intervention, whether or not considered related to it; the term encompasses any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the intervention. Clinical adverse events are graded for severity by the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. In addition, a Physical AI adverse event, per 21 CFR §312.32(f), is any untoward occurrence associated with the operation of the device during the investigation, whether or not caused by it, including robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized system access, digital-twin divergence, and any event requiring activation of the emergency stop [1].

9.3.2 Definition of Serious Adverse Events (SAE)

A serious adverse event is any adverse event that results in death, is life-threatening, requires inpatient hospitalization or prolongation of existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is an important medical event that may jeopardize the participant and may require intervention to prevent one of the foregoing outcomes. A serious Physical AI adverse event, per §312.32(f), is a Physical AI adverse event that results in, or has the potential to result in, any serious-adverse-event outcome, or that results in an unplanned interruption of an investigational drug-administration procedure requiring clinical intervention, an uncontrolled release of the investigational drug, or compromise of the device's sterile field during a surgical procedure.

9.3.3 Classification of an Adverse Event

Each adverse event is classified along four axes. Severity is graded by CTCAE version 5.0 (Grade 1 mild through Grade 5 fatal). Causality is assessed separately in relation to the drug and in relation to the device, each as related or unrelated (with relatedness denoting a reasonable possibility that the component caused the event), so that a single event may carry independent drug and device causality attributions. Expectedness is judged against the reference safety information (the daraxonrasib investigator's brochure for the drug and the Physical AI supplement and System Description for the device), and an event not listed there at the observed specificity or severity is classified as unexpected. Seriousness is determined against the criteria in §8.3.2. The combination of unexpected, serious, and at least possibly related defines a suspected adverse reaction reportable on the expedited timelines below.

9.3.4 Time Period and Frequency for Assessment and Follow-Up

Adverse events are solicited and recorded from the time of informed consent and are assessed at every study contact. The primary intensive safety-collection window spans from consent through the 30-day postoperative safety visit and encompasses the 28-day daraxonrasib DLT window. Beyond the 30-day window, serious adverse events, events of special interest, Physical AI adverse events, and new anti-cancer therapy continue to be collected through the long-term follow-up period to the 24-month overall-survival endpoint. Each event is followed until resolution, stabilization, return to baseline, or a determination that no further change is expected, or until the participant is lost to follow-up or withdraws consent.

9.3.5 Adverse Event Reporting

All adverse events, serious and non-serious, are recorded in the case report form with onset and resolution dates, severity grade, drug and device causality, expectedness, seriousness, action taken with each intervention, and outcome. Routine (non-serious) events are captured on the adverse-event case report form at the scheduled visits and do not require expedited transmission. Device-telemetry-derived observations that meet the adverse-event definition are entered with reference to the corresponding audit-trail record.

9.3.6 Serious Adverse Event Reporting

Serious adverse events are reported by the investigator to the sponsor within 24 hours of awareness. The sponsor submits IND/IDE safety reports to the FDA on the regulatory timelines: any unexpected fatal or life-threatening suspected adverse reaction is reported as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable. In addition to the clinical SAE pathway, six Physical AI reporting triggers are reported under §312.32(g): (i) any serious Physical AI adverse event, on the 7- or 15-day timeline as applicable; (ii) any Physical AI system-drug interaction event causing incorrect drug administration (wrong dose, route, timing, or patient), regardless of seriousness; (iii) any cybersecurity incident, within 15 calendar days (or 7 days if it has the potential to cause patient harm); (iv) sustained AI model performance degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; (v) any sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation > 4 mm or force discrepancy > 1.0 N); and (vi) any digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical AI adverse event reported, the complete hash-chained audit trail covering the period from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11 and made available to the FDA on request [19].

9.3.7 Events of Special Interest

The following events are designated events of special interest and are tracked and reported with the same diligence as serious adverse events regardless of whether they independently meet seriousness criteria: vascular injury (including any hard-stop E-stop precipitated by a vascular safety-zone breach), clinically relevant postoperative pancreatic fistula (ISGPS Grade B or C), conversion from device-directed to manual or open technique, any cybersecurity incident affecting the device, and any model-performance degradation. Each is reviewed by the Physical AI Safety Review Committee, which convenes at intervals not exceeding 90 days during active enrollment, and is reported to the data safety monitoring board.

9.3.8 Reporting of Pregnancy

Pregnancy is not an adverse event but is reported because of the potential exposure of an embryo or fetus to daraxonrasib. Participants of childbearing potential and partners of participants must use highly effective contraception during treatment and for the protocol-specified interval afterward. Any pregnancy in a participant or a participant's partner occurring during treatment or within that interval is reported by the investigator to the sponsor within 24 hours of awareness and is followed to outcome; a pregnancy with an associated serious adverse event, or an abnormal pregnancy outcome, is reported on the applicable expedited SAE timeline.

9.4 Unanticipated Problems

9.4.1 Definition

An unanticipated problem is any incident, experience, or outcome that is unexpected (not described in the protocol, consent, investigator's brochure, or Physical AI System Description in terms of nature, severity, or frequency given the procedures and population), related or possibly related to participation, and that suggests the research places participants or others at a greater risk of harm (including physical, psychological, economic, or social harm) than was previously known or recognized. For the device, the Physical AI hold grounds (repeated uncorrected safety failures, a confirmed cybersecurity compromise of drug-administration parameters or system integrity, and a simulation-validation deficiency that undermines the pre-clinical safety basis) are treated as paradigm unanticipated problems.

9.4.2 Reporting

Unanticipated problems are reported by the investigator to the reviewing IRB and, as applicable, to the sponsor and the FDA in accordance with the institutional policy and the IND/IDE requirements. An unanticipated problem that is also a serious and unexpected suspected adverse reaction is reported on the 7- or 15-day timeline of §8.3.6; other unanticipated problems are reported promptly and ordinarily within the institution's defined window (typically no later than the next scheduled continuing review, and sooner when the problem indicates an immediate hazard). Reports that involve the device preserve the hash-chained audit trail across the event window. The Physical AI Safety Review Committee and the data safety monitoring board review each unanticipated problem and recommend any change to device operation, human-oversight requirements, USL thresholds, or the protocol.

9.4.3 Reporting to Participants

When an unanticipated problem affects the rights, safety, or welfare of enrolled participants, or could affect a participant's willingness to continue, the information is communicated to affected participants in a timely manner, ordinarily through an IRB-approved re-consent or written notification. Where an unanticipated problem changes the risk profile of the combined intervention, including a relevant Physical AI safety event, the consent document is updated and currently enrolled participants are re-consented before any further study procedure, consistent with the Physical AI opt-out carried in the original consent.

10 Statistical Considerations

The statistical framework for this study is that of a small, intensively monitored, single-arm, first-in-human investigation in which the primary purpose is to characterize early safety and to identify a recommended dose, not to test a comparative efficacy hypothesis. Accordingly, the analyses are predominantly descriptive and estimation-based rather than confirmatory, and the design is governed by its operating characteristics rather than by formal statistical power. All analyses follow the principles of the International Council for Harmonisation (ICH) E9 guideline on statistical principles for clinical trials, are prespecified in this section and in the Statistical Analysis Plan (SAP) that this section summarizes, and are reported in accordance with the TRIPOD+AI statement for the model-assisted advisory component [20].

10.1 Statistical Hypotheses

The study has one primary safety hypothesis and one dose-finding objective, each matched to an analytic strategy appropriate for the small sample.

The primary safety hypothesis is framed as a one-sided assessment against a prespecified unacceptable threshold. Let π denote the true incidence of device-related or procedure-related serious adverse events (serious AEs) within the 30-day perioperative window. The null and alternative hypotheses are $H_0 : \pi \geq \pi_0$ versus $H_1 : \pi < \pi_0$, where π_0 is the prespecified unacceptable serious-AE rate. The trial is designed so that a device-related serious-AE rate at or above the cohort-level halt threshold defined in §9.4.6 (a rate $\geq$ one third within a cohort, or two device-related deaths) triggers a mandatory data and safety monitoring board (DSMB) review and a possible halt. Because of the small sample, this hypothesis is evaluated by estimation: the observed serious-AE proportion is reported with an exact (Clopper-Pearson) two-sided 95% confidence interval, and acceptability is judged by whether the interval excludes, or fails to exclude, the unacceptable threshold π_0, rather than by a fixed-rejection significance test that a single-arm safety cohort cannot reliably support.

The dose-finding objective is to identify the maximum tolerated dose (MTD) and the recommended Phase 2 dose (RP2D) of perioperative daraxonrasib using the standard 3+3 rule (§4.1, §6). This objective is operational rather than hypothesis-testing: the MTD is the highest dose level at which fewer than two of up to six participants experience a dose-limiting toxicity (DLT), and the RP2D is selected by integrating the MTD with the totality of the safety, tolerability, telemetry, and available pharmacodynamic information. No formal type I error is spent on dose selection; the operating characteristics of the 3+3 rule (§9.2) govern the dose-finding behavior. ICH E9 estimands principles are applied to the supportive efficacy endpoints: the population of interest, the endpoint, the handling of intercurrent events (for example open conversion, withdrawal, or death), and the population-level summary are each specified for every endpoint in §9.4.

10.2 Sample Size Determination

The planned sample size is up to $n = 18$ treated participants, allocated across three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule. This number is not derived from a power calculation, because the study tests no confirmatory efficacy hypothesis; it is determined jointly by the dose-finding design and by the early-feasibility precedent for significant-risk devices, and it is justified by its operating characteristics.

The 3+3 rule enrolls cohorts of three. A dose level that yields no DLT in three participants escalates; a dose level that yields one DLT in three expands to six, and a dose level that yields two or more DLTs in up to six is exceeded, defining the dose below it as the MTD. The maximum exposure at any single dose level is therefore six participants, and three escalation levels give a design maximum of $3 \times 6 = 18$ treated participants; many realizations terminate with fewer. This cohort size is consistent with the FDA early-feasibility guidance for significant-risk devices, under which limited initial clinical experience, generally fewer than ten participants per device-feasibility cohort, is sought when nonclinical simulation alone cannot supply the information needed to advance development [17]. The combined ceiling of 18 thus respects the early-feasibility convention at the device-cohort level while providing the per-dose exposure that the drug-arm 3+3 escalation requires.

Rather than power, the design is characterized by its operating characteristics. The probability that the 3+3 rule escalates past a given dose level is a function of the true DLT probability at that level: a dose with a true DLT rate of roughly 10% is very likely to be tolerated and escalated, whereas a dose with a true DLT rate of 40% or higher is very likely to be identified as excessively toxic and to define the MTD at the level below. Precision of estimation at the planned sample is summarized in Table 3: with n between 3 and 18, an observed proportion is reported with an exact 95% confidence interval whose width reflects the small sample, so that, for example, zero events in 18 participants yields an exact upper 95% confidence bound near 18%, and one event in 18 yields an interval excluding rates above roughly one quarter. No formal significance level is set for the dose-finding analysis; for the supportive estimation analyses, two-sided 95% confidence intervals are reported throughout (nominal $\alpha = 0.05$, two-sided), and no multiplicity adjustment is applied because these analyses are descriptive and non-confirmatory.

Table 3: Operating characteristics and exact-interval precision at the planned sample sizes. Confidence intervals are two-sided exact (Clopper-Pearson) bounds for a single proportion; the dose-finding behavior is that of the standard 3+3 rule. No statistical power is computed because the study tests no confirmatory efficacy hypothesis.

Quantity	Planned value	Operating characteristic / precision
Cohort size	3, expand to 6	3+3 rule: escalate on 0/3; expand on 1/3; exceed dose on $\geq 2/6$.
Per-dose maximum	6	Maximum exposure at any one dose level before the MTD is declared.
Dose levels	3 (DL1 to DL3)	Three-level escalation; design maximum $3 \times 6 = 18$ treated.
Total treated	up to 18	At or below the early-feasibility convention of generally fewer than 10 per device cohort, summed across dose levels [17].
0 events / 3	point estimate 0%	Exact two-sided 95% CI approximately 0% to 71%.
0 events / 6	point estimate 0%	Exact two-sided 95% CI approximately 0% to 46%.
0 events / 18	point estimate 0%	Exact two-sided 95% CI approximately 0% to 18%.
1 event / 18	point estimate 5.6%	Exact two-sided 95% CI approximately 0.1% to 27%.
Significance level	$\alpha = 0.05$ (two-sided)	Applies to supportive estimation only; no α is spent on dose selection.

10.3 Populations for Analyses

Four analysis populations are defined and applied consistently across all analyses. The Safety population comprises all participants who received any study intervention (any dose of daraxonrasib or any patient-facing robotic activity) and is the primary population for all safety and primary-endpoint analyses. The DLT-evaluable population comprises all dosed participants who either completed the 28-day DLT-evaluation window or experienced a DLT within it, and is the population on which the 3+3 escalation decisions and the MTD determination are based. The Per-Protocol population comprises participants who received the robotic Whipple and the LLM-directed advisory per protocol without a major protocol deviation that could affect the efficacy assessment, and supports the supportive surgical-quality analyses. The modified intention-to-treat (mITT) population comprises all dosed participants who contributed at least one post-baseline assessment, and is used for the supportive and exploratory efficacy summaries. Figure 22 shows the derivation of the four populations and the cohort-level interim stopping logic.

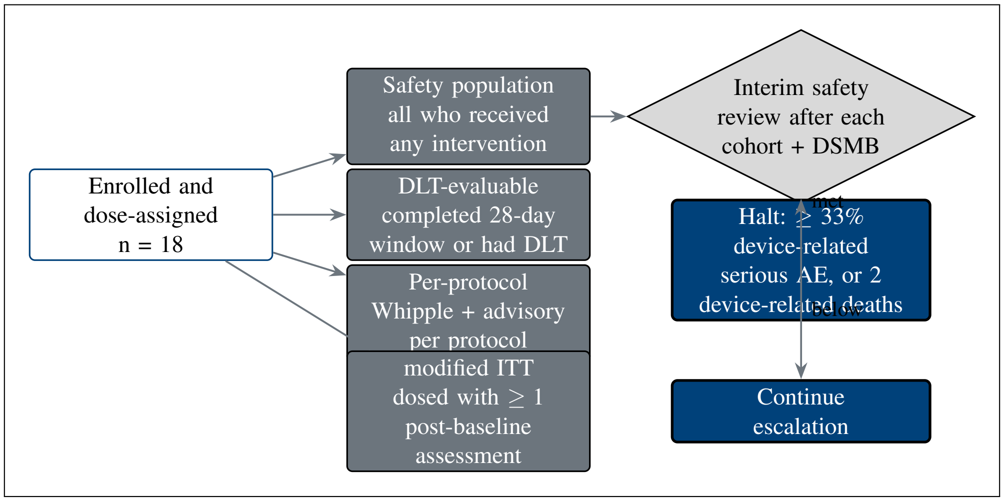

Figure 22. The four analysis populations (safety, dose-limiting-toxicity evaluable, per-protocol, modified intention-to-treat) and the cohort-level interim safety review with prespecified DSMB halt rules (a device-related serious-AE rate at or above one third within a cohort, or two device-related deaths).

10.4 Statistical Analyses

10.4.1 General Approach

All analyses are descriptive and estimation-based. Continuous variables are summarized by the number of non-missing observations, mean, standard deviation, median, minimum, and maximum; categorical variables are summarized by counts and percentages. Because the sample is small, every proportion (incidence, rate, or fraction) is accompanied by an exact (Clopper-Pearson) two-sided 95% confidence interval rather than a normal-approximation (Wald) interval, which is unreliable near the bounds and at small n. Time-to-event variables are summarized by the Kaplan-Meier method with the number at risk, and no inferential model is fitted unless prespecified as exploratory. Missing data are not imputed for the primary or safety analyses; the analysis population denominators are reported explicitly, and missingness is described. No interim or final hypothesis test is conducted for efficacy, so no multiplicity adjustment is required; the few inferential summaries that appear are flagged as exploratory and non-confirmatory. All analyses are performed in a validated, version-controlled statistical environment under a deterministic random seed, and the analysis code and derived data sets are deposited so that every reported figure is reproducible.

10.4.2 Analysis of the Primary Endpoint(s)

The co-primary safety and feasibility endpoints are analyzed on the Safety population. The primary safety endpoint, the incidence of device-related or procedure-related serious AEs within 30 days, is reported as a proportion with an exact two-sided 95% confidence interval, overall and by dose level, and the upper confidence bound is compared against the prespecified unacceptable threshold π_0 (§9.1). The dose-finding endpoint, the MTD and the RP2D, is determined on the DLT-evaluable population by applying the 3+3 rule to the observed DLT counts at each dose level; the MTD is reported as the highest dose with fewer than two DLTs in up to six participants, and the RP2D is reported with the rationale that integrated the MTD with the safety, tolerability, and telemetry data. The operative feasibility endpoint, the proportion of cases in which the assigned operative tasks are completed under the staged-autonomy model without conversion attributable to system failure, is reported as a proportion with an exact 95% confidence interval. No formal significance test is applied to any primary endpoint; acceptability is judged by the confidence intervals and the prespecified thresholds.

10.4.3 Analysis of the Secondary Endpoint(s)

Secondary surgical-quality and short-term oncologic endpoints are analyzed on the Per-Protocol population (with the Safety population as a sensitivity analysis) and each is reported as a proportion with an exact two-sided 95% confidence interval, accompanied by the contemporaneous Dutch 2025 nationwide robotic-pancreatoduodenectomy benchmark as a descriptive, non-inferential comparator [5]. The prespecified secondary endpoints and their comparators are summarized in Table 4. The margin-negative (R0) resection rate, the rate of clinically relevant (ISGPS grade B or C) postoperative pancreatic fistula graded per the 2016 ISGPS update [4], the rate of Clavien-Dindo grade III or higher complications, and 30-day and 90-day mortality are each estimated with exact intervals. Because the comparison with the Dutch benchmark is descriptive and the study is single-arm, no hypothesis test of non-inferiority or superiority is performed; the benchmark is presented as context for interpreting the observed point estimates and intervals.

Table 4: Secondary surgical-quality and short-term oncologic endpoints, the analysis population, and the descriptive Dutch 2025 nationwide robotic-pancreatoduodenectomy comparator. Comparisons are descriptive only; the single-arm design supports estimation, not inferential between-group testing.

Secondary endpoint	Population / method	Reporting and comparator
Margin-negative (R0) resection rate	Per-protocol; exact 95% CI	Proportion with exact CI; descriptive comparison with the Dutch 2025 cohort [5].
Clinically relevant pancreatic fistula (ISGPS grade B/C)	Per-protocol; exact 95% CI	Graded per the 2016 ISGPS update [4]; benchmarked descriptively to Dutch 2025.
Clavien-Dindo grade III+ complications	Safety; exact 95% CI	Proportion with exact CI; Dutch 2025 descriptive comparator.
30-day mortality	Safety; exact 95% CI	Proportion with exact CI; Dutch 2025 descriptive comparator.
90-day mortality	Safety; exact 95% CI	Proportion with exact CI; Dutch 2025 descriptive comparator.
Estimated blood loss, operative time, length of stay	Per-protocol; descriptive	Mean / median with range; Dutch 2025 phase-of-learning-curve context.

10.4.4 Safety Analyses

Safety is the focus of the study, and all safety analyses are performed on the Safety population. Adverse events are coded to the Medical Dictionary for Regulatory Activities (MedDRA) and tabulated by system organ class and preferred term, by maximum severity using the Common Terminology Criteria for Adverse Events (CTCAE), and by relationship to the drug, to the device, and to the procedure. Serious AEs, DLTs, and deaths are listed individually. Treatment-emergent AEs are summarized overall and by dose level, with the number of participants reporting each event, the number of events, and the corresponding proportions with exact 95% confidence intervals.

Device telemetry constitutes a second, parallel safety stream unique to this trial class. The cyber-physical record is summarized per procedure and in aggregate: emergency-stop activations and the measured stop latency against the $\leq$ 500 ms requirement; positional and force-tolerance excursions against the $\leq$ 2 mm and $\leq$ 0.5 N envelopes; vascular-exclusion gating events; and the counts of LLM advisories that were accepted, blocked, or escalated to the human operator by the verification-before-generation gate surface. A dedicated Physical AI adverse-event tabulation records, for every event, whether an LLM advisory, a robot command, a safety-limit trip, or a human override was implicated, so that events arising from the autonomy-bearing layer are visible and traceable to a deterministic seed and a repository commit through the hash-chained audit trail (§10.3). Laboratory data, vital signs, and physical-examination findings are summarized descriptively with shift tables relative to baseline.

10.4.5 Baseline Descriptive Statistics

Baseline characteristics are summarized descriptively for each analysis population. Reported variables include demographics (age, sex, race, ethnicity); disease staging and resectability category (resectable versus borderline-resectable); the specific KRAS G12 mutation; baseline CA 19-9; Eastern Cooperative Oncology Group (ECOG) performance status; prior neoadjuvant therapy and number of cycles; and relevant comorbidity and organ-function indices. No formal between-group comparison of baseline characteristics is performed, consistent with the single-arm design; baseline balance across dose levels is described to support interpretation of the safety and dose-finding results.

10.4.6 Planned Interim Analyses

Formal interim safety analyses are conducted at the cohort level. After each participant completes the 28-day DLT-evaluation window and before the next participant in the cohort undergoes surgery, the accumulating safety data are reviewed under the staggered (sentinel) enrollment rule (§4.1); and after each dose-level cohort is complete, the DSMB convenes for a formal interim safety review and an escalation, expansion, hold, or halt decision. The prespecified halt rules are quantitative and binding: enrollment and escalation halt for a mandatory DSMB review if the device-related or procedure-related serious-AE rate reaches or exceeds one third within a cohort, or if two device-related deaths occur at any point. Additional pause triggers include any unexpected device-related serious AE, any breach of a hard cyber-physical safety limit (emergency-stop latency, positional, or force envelope), and any failure of the pre-procedure safety-matrix verification. The DSMB may halt, pause, modify, or permit continuation; a halt for an immediate hazard is implemented at once under §312.30(h) and §10.10. No interim efficacy analysis and no efficacy-based stopping rule are specified, so no efficacy-related type I error is spent. The stopping logic is shown in Figure 22.

10.4.7 Sub-Group Analyses

Prespecified subgroup summaries are descriptive and hypothesis-generating only. Safety and the supportive surgical-quality endpoints are summarized by resectability category (resectable versus borderline-resectable) and by daraxonrasib dose level (DL1 through DL3). Given the small sample, subgroup results are reported as counts and proportions with exact confidence intervals and are interpreted with explicit caution; no formal test of interaction or of subgroup difference is performed, and no clinical conclusion is drawn from any single subgroup.

10.4.8 Tabulation of Individual Participant Data

Because the cohort is small and intensively monitored, individual participant data are tabulated in full. Participant-level listings are provided for demographics and baseline characteristics; dose level and actual exposure; all adverse events with onset, severity, seriousness, relationship, and outcome; DLTs and the resulting escalation decisions; serious AEs and deaths; surgical-quality outcomes; and the per-procedure device-telemetry and Physical AI event summaries, each keyed to participant and timepoint. Listings are presented so that every aggregate statistic can be traced to its contributing records, consistent with the transparency expected of a first-in-human cyber-physical investigation.

10.4.9 Exploratory Analyses

Exploratory analyses are non-confirmatory and are clearly labeled as such. Progression-free survival (PFS) and overall survival (OS) through the 24-month follow-up are summarized by the Kaplan-Meier method with the number at risk, and are interpreted as descriptive given the sample and the absence of a comparator. The concordance between LLM advisories and the actions ultimately taken by the operating surgeon, and the sim-to-real gap between the Phase 0 simulation predictions and the observed intraoperative telemetry, are summarized descriptively to inform later-phase design and the verification-before-generation calibration. These exploratory results generate hypotheses for a later-phase study and are not used to draw efficacy conclusions in this protocol.

11 Regulatory, Ethical, and Oversight Considerations

The oversight architecture for this combined IND/IDE first-in-human study layers a conventional human-subject-protection structure with the additional governance that an autonomy-bearing surgical system requires. The Sponsor-Investigator, ChemicalQDevice, answers to a reviewing Institutional Review Board (IRB) and to an independent data and safety monitoring board (DSMB), convenes a Physical AI Safety Review Committee on a $\leq$ 90-day cadence, designates an Independent Safety Monitor (ISM) for each procedure, and delegates day-to-day monitoring to a robotics- and artificial-intelligence-competent clinical research organization (CRO) over the site team. Figure 16 presents the governance and safety-oversight structure.

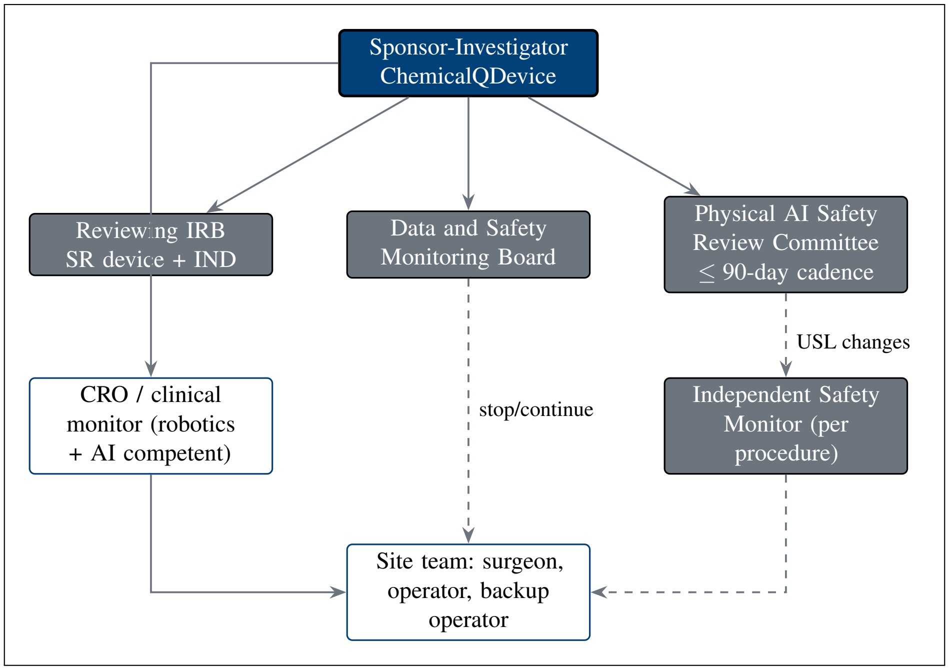

Figure 16. Study governance and safety oversight. The Sponsor-Investigator answers to the reviewing IRB and the DSMB, convenes a Physical AI Safety Review Committee on a $\leq$ 90-day cadence, designates an Independent Safety Monitor for each procedure, and delegates monitoring to a robotics- and AI-competent CRO over the site team (surgeon, operator, and a qualified backup operator).

11.1 Informed Consent Process

Informed consent is obtained by a qualified member of the research team, who is not the treating surgeon, before any study-specific procedure, in language the participant can understand and with adequate time and opportunity to ask questions, consistent with 21 CFR part 50 [21] and ICH E6(R3). The consent discussion covers the diagnosis, the alternatives, and the prognosis; the perioperative daraxonrasib regimen and its risks; and, in addition to the standard elements, an explicit and distinct disclosure of the on-premises LLM-directed robotic system, its role and continuous human oversight, and its system-specific risks, as required for an autonomy-bearing investigational device. The consent form addresses the therapeutic misconception directly and explains the staged-autonomy model under which the system operates only at the clinician-controlled and supervised semiautonomous levels.

A defining feature of this trial class is the documented Physical AI opt-out. Under 21 CFR §312.60(f), as adapted for Physical AI investigations [1], the participant is offered, and may exercise, a documented right to request administration without the Physical AI advisory layer where clinically feasible, without penalty and without affecting eligibility for the remainder of the study or for clinical care. The participant's election or declination of the opt-out is recorded in the source document and the case report form (CRF). Assent is obtained where applicable, although the adult-only eligibility makes pediatric assent inapplicable in practice; a legally authorized representative process and the use of qualified interpreters and translated materials for participants with limited English proficiency are provided so that consent is meaningful for every eligible individual. The right to withdraw at any time without penalty is stated at consent and reaffirmed at each contact. Figure 17 shows the consent process and the opt-out.

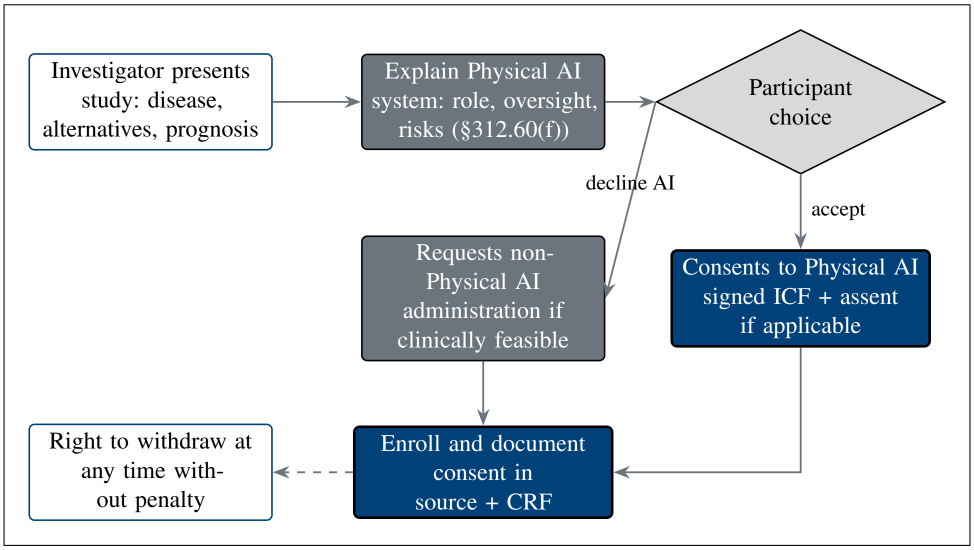

Figure 17. Informed-consent process with the Physical AI opt-out. Consent discloses the Physical AI system's role, continuous human oversight, and system-specific risks, and offers the participant a documented right to request non-Physical AI administration where clinically feasible (21 CFR §312.60(f)), with the standard right to withdraw at any time without penalty.

11.2 Study Discontinuation and Closure

The study may be discontinued or closed by the Sponsor-Investigator, by the IRB, or by the FDA. Grounds for closure include unacceptable safety findings (including the prespecified halt rules of §9.4.6), a clinical hold, a determination that the study cannot be conducted in compliance, or completion of the planned cohorts. The FDA may impose a clinical hold on the drug or device component under the provisions for terminating or suspending an investigation; in addition, the Physical AI overlay establishes specific clinical-hold grounds for the autonomy-bearing system, including a breach of a hard cyber-physical safety limit, a failure of the verification-before-generation gate surface, or a loss of audit integrity. On any hold or closure, enrollment and patient-facing robotic activity stop immediately; affected participants are transitioned to appropriate clinical care; and a controlled decommissioning and data-preservation procedure is executed, under which the robotic system is locked out of patient contact, the deterministic configuration and repository commit are recorded, and the complete hash-chained command and telemetry record is sealed and retained for the regulatory retention period (§10.9). The IRB, the FDA, and the DSMB are notified, and a final report is prepared.

11.3 Confidentiality and Privacy

Participant confidentiality is protected throughout. Data released for analysis, deposition, or publication are de-identified by the HIPAA Safe Harbor method, with removal of all eighteen categories of protected health information identifiers (names; geographic subdivisions smaller than a state; all date elements other than year that are directly related to an individual, and ages over 89; telephone and fax numbers; email addresses; Social Security numbers; medical record numbers; health plan beneficiary numbers; account numbers; certificate or license numbers; vehicle identifiers; device identifiers and serial numbers; web URLs; IP addresses; biometric identifiers; full-face photographs and comparable images; and any other unique identifying number, characteristic, or code). Access to identifiable records is governed by a deny-by-default authorization model, in which no role can read, write, or export a record except where an access right has been explicitly granted, and every access, change, and export is written to a hash-chained audit trail in which each entry cryptographically incorporates the hash of the prior entry so that the record is tamper-evident and any alteration is detectable. All electronic records, signatures, and the cyber-physical telemetry stream are maintained under 21 CFR part 11 [19], with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate.

11.4 Future Use of Stored Specimens and Data

The cyber-physical record is retained as a tiered telemetry pyramid that balances long-term scientific reuse against storage cost. At the apex are the compact, human-readable summaries and the safety-event and audit logs retained in full and indefinitely; the middle tiers hold compressed feature and event streams; and the base, level zero (L0), holds the raw, losslessly reconstructable command and sensor stream. The L0 raw tier is deposited on Zenodo under controlled terms and is decompressible and reconstructable on FDA request as required for inspection and review under 21 CFR §312.57, so that any reported behavior can be replayed deterministically from the seed and the repository commit. Any future secondary use of stored de-identified data is limited to research consistent with the consent and applicable regulation, is conducted on de-identified data only, and preserves the audit chain. No biological specimens are banked beyond those required for the clinical care and the protocol-specified assessments; tumor genotyping is performed on standard-of-care material, and no specimen is retained for an unspecified future purpose without appropriate consent.

11.5 Key Roles and Study Governance

The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak) holds the combined IND/IDE and bears the sponsor and investigator responsibilities under 21 CFR parts 312 and 812, including protocol custody, safety reporting, oversight of the investigation, and assurance of compliance. The site Principal Investigator directs the conduct of the study at the clinical site and supervises the research team. The operating surgeon performs the pancreaticoduodenectomy under the staged-autonomy model; a primary operator drives the system, and a qualified backup operator is present and ready to assume control or to invoke the manual override at any time. The CRO provides robotics- and AI-competent clinical monitoring. The DSMB, the Independent Safety Monitor, and the Physical AI Safety Review Committee provide the independent oversight described in §10.6. Roles, delegated duties, and the corresponding qualifications are documented on a delegation-of-authority log maintained by the site, and changes are recorded and reported as required.

11.6 Safety Oversight

Independent safety oversight operates on three tiers. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, reviews accumulating safety data at each cohort boundary and at any triggered review, applies the prespecified halt rules of §9.4.6, and recommends continuation, modification, pause, or halt. The Independent Safety Monitor is designated for each procedure and provides real-time, case-level safety oversight independent of the operating team, with the authority to call a stop. The Physical AI Safety Review Committee meets on a $\leq$ 90-day cadence (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the system: the telemetry and audit record, the verification-before-generation gate-surface decisions, the Unified Safety Level (USL) rating, and any software change, and to recommend USL re-assessment or oversight changes to the Sponsor-Investigator. The three tiers are coordinated so that no autonomy-bearing change, and no escalation decision, proceeds without the appropriate independent review.

11.7 Clinical Monitoring

The study is monitored by a CRO whose monitors are competent in both clinical-trial monitoring and the robotics and artificial-intelligence elements of this trial class. Monitoring follows a risk-based plan and includes source-data verification not only of conventional clinical source documents and the consent records but also of the cyber-physical evidence: the device telemetry, the verification-before-generation gate-surface logs, and the hash-chained audit trail, which are verified against the recorded outcomes and the deterministic seed and repository commit. Monitoring visits verify eligibility, consent (including the documented Physical AI opt-out election), safety reporting, protocol adherence, and data integrity, and findings are documented and resolved through to closure.

11.8 Quality Assurance and Quality Control

Quality assurance and quality control span the clinical and the cyber-physical domains. Before every procedure, the pre-procedure safety matrix is verified, the USL readiness rating is confirmed at or above the deployment threshold, and the emergency-stop, positional, and force envelopes are checked. After any software update, model change, or configuration change to the LLM-directed system, a Phase 0 re-validation is performed against the independent simulation frameworks, and the USL is reassessed, before any further patient-facing use; no update reaches a participant without re-validation. Audits may be conducted by the Sponsor-Investigator or a designee independent of the conduct of the study, and corrective and preventive actions are documented. The deterministic, version-controlled build ensures that the validated configuration in use is exactly the configuration that was tested and recorded.

11.9 Data Handling and Record Keeping

Seven Physical AI record types are captured and retained for this trial class, in addition to the conventional clinical trial records: (1) the LLM advisory record (each prompt, context, and advisory output); (2) the robot command record (each command issued to the arms); (3) the sensor and telemetry record (the cyber-physical stream, tiered as in §10.4); (4) the safety-limit-event record (emergency-stop activations and latencies, positional and force-envelope excursions, and vascular-exclusion gating events); (5) the verification-before-generation gate-surface record (each accept, block, or escalate decision); (6) the human-override and intervention record; and (7) the hash-chained audit trail that binds the preceding six to a deterministic seed and a repository commit. All records are retained for the period required by 21 CFR §312.57 (at least two years after a marketing application is approved for the indication, or, if no application is filed or approved, two years after the investigation is discontinued and the FDA is notified), and the L0 raw tier is preserved as described in §10.4 so that any record can be reconstructed on FDA request.

11.10 Protocol Deviations

A protocol deviation is any departure from the IRB-approved protocol. Deviations are identified, documented, assessed for their effect on participant safety and data integrity, and reported to the IRB and the Sponsor-Investigator in accordance with the IRB's policy and applicable regulation; deviations that affect safety or the rights and welfare of participants, and any that recur, are escalated. Deviations are distinguished from planned protocol amendments, which follow the amendment process. Where a deviation is necessary to eliminate an apparent immediate hazard to a participant, the change may be implemented without prior IRB or FDA approval under 21 CFR §312.30(h) (and the corresponding device provision), with prompt notification and reporting afterward; for the autonomy-bearing system, an immediate-hazard modification also triggers the decommissioning and re-validation controls of §10.2 and §10.8 before any further patient-facing use.

11.11 Publication and Data Sharing Policy

Results are disseminated regardless of outcome through dual deposition: the analysis code, derived data sets, and protocol are published in a public GitHub repository and archived on Zenodo with a citable DOI, under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. Reporting follows the TRIPOD+AI statement for the model-assisted advisory component [20] and the CREMLS (Consensus Reporting for Evaluation of Machine Learning Systems) guidance for the machine-learning system, in addition to the applicable clinical-trial reporting standards. The deterministic seed and the repository commit are published so that every reported result is reproducible, and the tiered telemetry record is deposited as described in §10.4. No identifiable participant information is published; all shared data are de-identified by the HIPAA Safe Harbor method (§10.3).

11.12 Conflict of Interest Policy

Financial interests are disclosed and managed in accordance with 21 CFR part 54 (Financial Disclosure by Clinical Investigators). The Sponsor-Investigator is the chief executive of ChemicalQDevice, which develops the Physical AI system under study; this relationship is disclosed in full, and the independent oversight structure of §10.6 (the DSMB, the Independent Safety Monitor, and the Physical AI Safety Review Committee) is established in part to manage it, so that safety and escalation decisions are not made by a party with a financial interest in the outcome alone. Investigator financial-disclosure certifications are collected before participation and updated through one year after the study, and any identified conflict is disclosed and managed before it can affect the conduct, analysis, or reporting of the study.

12 Additional Considerations, Abbreviations, and Amendment History

12.1 Additional Considerations

Although this protocol is written to the United States combined IND/IDE pathway, the LLM-directed robotic system is a Software as a Medical Device (SaMD) whose deployment may be sought across jurisdictions, and the design anticipates the parallel regulatory tracks summarized in Table 5. The European Union Medical Device Regulation (EU MDR) and its companion artificial-intelligence requirements, China's National Medical Products Administration (NMPA), Japan's Pharmaceuticals and Medical Devices Agency (PMDA), the United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA), Health Canada, and Australia's Therapeutic Goods Administration (TGA) each maintain a SaMD and robotically assisted surgery framework with which the device evidence is designed to remain compatible. The common technical spine, the Phase 0 simulation validation, the USL readiness rating, the hash-chained audit trail, and the verification-before-generation gate surface, is constructed to map onto each track without redesign, and the international consensus standards cited in §12 (IEC 80601-2-77, ASME V and V 40, NASA-STD-7009A, and IEEE 7009-2024) provide the shared basis across jurisdictions.

The trial is additionally aligned with the legislative and financial framework of the Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510) [7], which ties the funding and oversight of Physical AI oncology investigations to a verification-and-validation-and-uncertainty-quantification (VVUQ) evidentiary standard. The ten-gate assurance suite operationalizes that standard: every proposed robot-patient command is verified before it is generated, against fourteen external safety and robotics standards plus two clinical baselines, with epistemic and aleatory uncertainty bounds, and the gate surface returns one of accept, block, or escalate. Figure 12 renders the ten-gate pipeline.

Table 5: Cross-jurisdictional Software as a Medical Device (SaMD) and robotically assisted surgery tracks with which the device evidence is designed to remain compatible. The common technical spine (Phase 0 validation, USL readiness, the hash-chained audit trail, and the verification-before-generation gate surface) maps onto each track without redesign.

Jurisdiction	Regulator / framework	Alignment basis
European Union	EU MDR + AI requirements	SaMD risk classification; common simulation-validation and audit evidence.
China	NMPA	SaMD and robotically assisted surgery review; standards-based VVUQ dossier.
Japan	PMDA	SaMD and device review; consensus-standard alignment.
United Kingdom	MHRA	SaMD framework; post-market and audit-trail compatibility.
Canada	Health Canada	SaMD licensing; quality-system and validation evidence.
Australia	TGA	SaMD inclusion; standards-based safety and performance evidence.

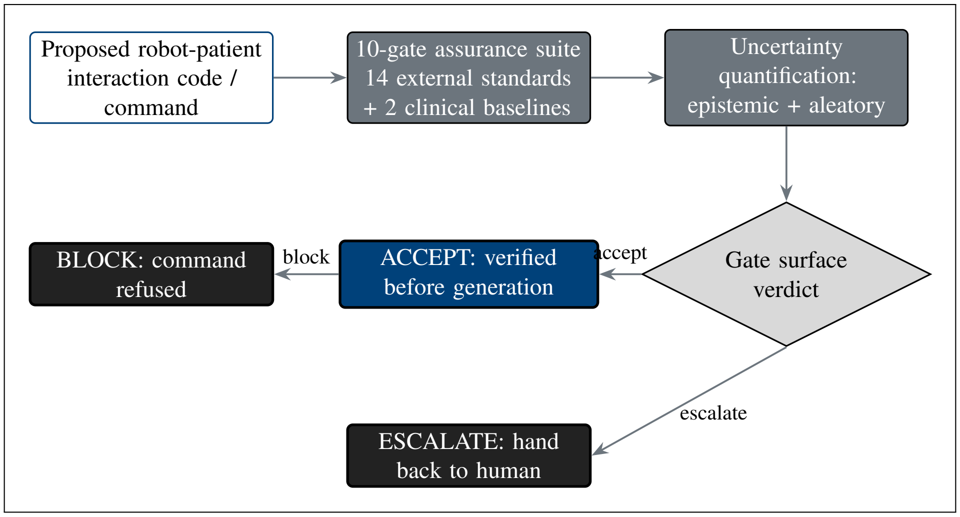

Figure 12. Verification before generation. Every proposed robot-patient command passes a ten-gate assurance suite aligned to fourteen external safety and robotics standards plus two clinical baselines, with epistemic and aleatory uncertainty bounds, before the gate surface returns ACCEPT, BLOCK, or ESCALATE (hand back to human). Prior author suites cleared 51 of 51 and 172 of 172 tests.

12.2 Abbreviations

Table 6 resolves every abbreviation used across this protocol.

Table 6: Abbreviations used in this protocol.

Abbr.	Definition	Abbr.	Definition
AE	Adverse event	MCP	Model Context Protocol
CTCAE	Common Terminology Criteria for Adverse Events	MTD	Maximum tolerated dose
DLT	Dose-limiting toxicity	OS	Overall survival
DSMB	Data and Safety Monitoring Board	PDAC	Pancreatic ductal adenocarcinoma
ECOG	Eastern Cooperative Oncology Group	PFS	Progression-free survival
FDA	U.S. Food and Drug Administration	PJ	Pancreaticojejunostomy
GCP	Good Clinical Practice	PV	Portal vein
GJ	Gastrojejunostomy	R0	Margin-negative resection
HJ	Hepaticojejunostomy	RP2D	Recommended Phase 2 dose
ICH	International Council for Harmonisation	SAE	Serious adverse event
IDE	Investigational Device Exemption	SMA	Superior mesenteric artery
IND	Investigational New Drug	SMV	Superior mesenteric vein
ISGPS	International Study Group of Pancreatic Surgery	USL	Unified Safety Level
ISM	Independent Safety Monitor	VVUQ	Verification, validation, and uncertainty quantification
ITT	Intention-to-treat		
KRAS	Kirsten rat sarcoma viral oncogene homolog		
LLM	Large language model		

12.3 Protocol Amendment History

This is the initial version of the protocol. Table 7 records the version history; subsequent amendments will be appended with the version, date, sections affected, and a summary of the change.

Table 7: Protocol amendment history.

Version	Date	Sections affected	Summary of change
v4.0.0	June 20, 2026	All (initial release)	Initial version of the protocol. Establishes the combined IND/IDE design, the perioperative daraxonrasib 3+3 dose-finding arm, the on-premises LLM-directed eight-arm robotic Whipple device arm, the Physical AI Subpart J overlay, the statistical and oversight framework, and the back matter.

13 References and Back Matter

The reference list is rendered below through the `ieeetr` bibliography style from `references.bib`, and includes the five daraxonrasib program entries; the three main documents on which this protocol builds; the directly relevant author works; the clinical references; and the FDA guidance, the applicable 21 CFR parts (11, 50, and 812), and the IEC, ASME, NASA, and IEEE consensus standards together with the NIH protocol template. The three main documents are cited here with a brief explanation of the role each plays.

- The 2030 PDAC sixty-second robotic Whipple and daraxonrasib simulation [9] is the simulation source from which the clinical figures, the exemplar PAT-PDAC-0001, and the operative and telemetry behavior in this protocol derive; it supplies the in silico evidence that the early-feasibility device arm is built upon.
- The Adaption of 21 CFR Part 312 for end-to-end Physical AI oncology trial unification [1] is the regulatory source for the Physical AI overlay, including Subpart J, the §312.60(f) Physical AI consent and opt-out, the Phase 0 simulation-validation gate, the USL readiness rating, and the hash-chained audit-trail requirements adopted throughout.
- The H. R. 9510 bill, Verification Before Generation in Physical AI Oncology Trials Act of 2026 [7], is the legislative and financial source that ties the funding and oversight of Physical AI oncology investigations to the VVUQ evidentiary standard operationalized by the ten-gate assurance suite (§11.1).

Acknowledgments

This protocol was assembled with Claude Code (Opus 4.8, 1M context). The author thanks the open-source robotics, simulation, and clinical-standards communities.

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Ethical Disclosures

This is an independent research draft. No real patients were enrolled, no protected health information was used, and no IRB approval was sought or obtained. The exemplar PAT-PDAC-0001 is synthetic and does not represent an enrolled individual. All clinical figures derive from the author's simulation sources and are illustrative unless tied to a cited reference. This document is not an active IND or IDE, is not an approved protocol, and is not medical or regulatory advice; it is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

Data Availability

The protocol, the analysis code, and the derived data sets are available through a public GitHub repository and archived on Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, at DOI 10.5281/zenodo.xxxxxxxx (the placeholder is filled at deposit). All deposited data are simulation-derived, generated under a deterministic seed, and contain no personally identifiable information.

Cite This Article

Kawchak K. A Phase 1, First-in-Human, Combined IND/IDE Clinical Trial Protocol of On-Premises Large Language Model Directed Robotic Pancreaticoduodenectomy (Whipple) with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

References

[1] K. Kawchak, "Adaption: 21 cfr part 312, end-to-end physical ai oncology clinical trial unification," Mar. 2026.

[2] R. L. Siegel, A. N. Giaquinto, and A. Jemal, "Cancer statistics, 2025," *CA: A Cancer Journal for Clinicians*, vol. 75, no. 1, pp. 10–45, 2025.

[3] Revolution Medicines, "Phase 3 study of daraxonrasib (rmc-6236) in previously treated metastatic pdac (rasolute 302)," 2025. NCT06625320.

[4] C. Bassi, G. Marchegiani, C. Dervenis, *et al.*, "The 2016 update of the isgps definition and grading of postoperative pancreatic fistula," *Surgery*, vol. 161, no. 3, pp. 584–591, 2017.

[5] Dutch Pancreatic Cancer Group, "Nationwide outcomes of 1000 robotic pancreatoduodenectomies across the four phases of the learning curve," *British Journal of Surgery*, 2025.

[6] T. Conroy *et al.*, "Folfirinox or gemcitabine as adjuvant therapy for pancreatic cancer," *New England Journal of Medicine*, vol. 379, no. 25, pp. 2395–2406, 2018.

[7] K. Kawchak, "Verification before generation in physical ai oncology trials act of 2026 (h. r. 9510), bill v5.0: A financial data amendment," June 2026.

[8] Revolution Medicines, "Fda breakthrough therapy designation for daraxonrasib in previously treated metastatic pancreatic cancer with kras g12 mutations," June 2025.

[9] K. Kawchak, "2030: 60 second pancreatic ductal adenocarcinoma robotic whipple procedure and daraxonrasib simulation," May 2026.

[10] K. Kawchak, "Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation," Oct. 2025.

[11] K. Kawchak, "Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook," Aug. 2025.

[12] K. Kawchak, "Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate," July 2025.

[13] K. Kawchak, "End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals," June 2025.

[14] R. Singh, M. Bapna, A. R. Diab, E. S. Ruiz, and W. Lotter, "How ai is used in fda-authorized medical devices: a taxonomy across 1,016 authorizations," *npj Digital Medicine*, vol. 8, p. 388, 2025.

[15] K. Kawchak, "Code generation competition: 16 proprietary vs. open-source llms and iterative learning based on fda adverse event reporting system," Dec. 2025.

[16] K. Kawchak, “Mobile pancreatic cancer unitree h2 surgical humanoid with priority vvuq,” May 2026.

[17] U.S. Food and Drug Administration, “Investigational device exemptions (ides) for early feasibility medical device clinical studies,” tech. rep., Center for Devices and Radiological Health, Oct. 2013.

[18] Electronic Code of Federal Regulations, “21 cfr part 812 - investigational device exemptions,” 2026.

[19] Electronic Code of Federal Regulations, “21 cfr part 11 - electronic records; electronic signatures,” 2026.

[20] G. S. Collins *et al.*, “Tripod+ai statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ*, vol. 385, p. e078378, 2024.

[21] Electronic Code of Federal Regulations, “21 cfr part 50 - protection of human subjects,” 2026.

[22] National Institutes of Health, “Nih-fda clinical trial protocol template,” 2017.

[23] International Electrotechnical Commission, “Iec 80601-2-77:2019 - basic safety and essential performance of robotically assisted surgical equipment,” 2019.

[24] ASME, “Asme v and v 40-2018 - assessing credibility of computational modeling through verification and validation: Application to medical devices,” 2018.

[25] National Aeronautics and Space Administration, “Nasa-std-7009a - standard for models and simulations,” 2016.

[26] IEEE, “Ieee std 7009-2024 - fail-safe design of autonomous and semi-autonomous systems,” 2024.